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**EXPLORING MODEL-AGNOSTIC INTERPRETABLE MACHINE
LEARNING FOR RED WINE QUALITY ASSESSMENT**

Name: Perla Pavan

Student ID: 24WJ1E00C5

Supervisor Name: Subramanyam sir

**FINAL PROJECT REPORT: EXPLORING MODEL-AGNOSTIC
INTERPRETABLE MACHINE LEARNING FOR RED WINE QUALITY
ASSESSMENT**

Abstract

This work addresses a critical challenge of organizing the wide variety of raw, chemical data into a machine learning model that is capable of explaining wine quality. Using techniques that are model agnostic like SHAP and LIME, this study offers quantitatively transparent and actionably insightful understanding into how chemical properties impact wine quality. The research systematically examines the essential factors: alcohol content, acidity and sugar level, and shows how artificial intelligence can provide objective quality evaluation beyond traditional subjective sensory assessment. The project has successfully developed a robust predictive model with 66 % accuracy and the Random Forest model performs well. The research provides winemakers with the ability to understand the exact chemical drivers of wine quality through innovative visualization techniques and interpretation frameworks. Beyond its immediate use for the wine industry, this study sets a precedence for AI in agricultural and food quality assessment and advocates for transparency and understanding in the drive towards technological innovation.

MSc FPR Declaration

This report is submitted in partial fulfilment of the requirement for the degree of: Master of Science in EXPLORING MODEL-AGNOSTIC INTERPRETABLE MACHINE LEARNING FOR RED WINE QUALITY ASSESSMENT, at the University of Hertfordshire (UH).

I hereby declare that the work presented in this project and report is my own, except where explicitly stated otherwise. All sources of information and ideas, whether quoted directly or paraphrased, have been properly referenced in accordance with academic standards. I understand that any failure to properly acknowledge the work of others could constitute plagiarism and may result in academic penalties.

I did not use human participants in my MSc Project.

I hereby give permission for the report to be made available on the university website provided the source is acknowledged.

Acknowledgment

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Chapter 1: Introduction to the Project

Background Information

Since the wine industry has traditionally relied on subjective sensory evaluation to judge wine quality, the assessments are not consistent. Thanks to advancements in technology, machine learning (ML) has proven to be an extremely helpful way of predicting wine quality using patterns of the chemical composition, which affects this quality. However, these models end up being 'black boxes' which prevents stakeholders, including winemakers, from trusting and interpreting their predictions. However, the difficulty of explaining predictions of ML models has not been addressed by model specific interpretability techniques. In this project, it explores their use for red wine quality prediction in order to design transparent, accurate, and practical prediction systems (Condran *et al.* 2022). The project helps winemakers understand how factors including acidity, alcohol content and sugar levels impact wine quality by bridging the gap between complex mathematical models and an application of those models to real world practice. Predicting quality is not enough, the goal is to be able to make these predictions actionable and reliable. It aligns with wider industry requirements for objective, trusted and comprehensible quality assessment tools in order to support improved decision making and provide a platform for innovation in wine production.

The next model's execution requires interpretable solutions modern ML models applied to quality assessment are complex and difficult to explain, which is critical when providing accurate and comprehensible quality forecasts for the wine sector (Cetin, 2022). More specifically, SHAP and LIME approaches are examined in assessment of red wine quality using interpretable machine learning methods. In this regard, the research aims at bridging the gap between complex mathematical models and practical usage by providing clear recommendations about how chemical properties of the components used in wine production relate to the wine quality. The objective of this project is to develop a solid model on wine quality output but also with an ability to extend a clear analytical explanation to the wine producers through an understanding of a complex wine data set containing large quantity of red wine samples with other features such as the alcohol quality, acidity, and the sugar content measurement (Carneiro *et al.* 2024). First, this study is significant as it may revolutionize the wine industry by proposing new objective and interpretable approaches that will bring a trustworthy external independent assessment of wine quality while maintaining audit trail. As it has been stipulated in the case of the food and the beverage industry where understanding the rationale behind quality prediction is as important as the prediction itself, this paper seeks to advance the use of explainable AI.

Project Aim

This project aims to develop an interpretable machine learning solution for assessing red wine quality, such that the gap between complex mathematical models and practical applications could be bridged. Its purpose is to provide transparent explanations while maintaining model accuracy and help wine producers understand how chemical properties influence wine quality through clear analytical explanations. The main idea of the project is to build an interpretable machine learning model to predict red wine quality using model agnostic techniques such as SHAP and LIME to make it transparent and a workable concept to the stakeholders.

Project Objective

- To implement and evaluate model-agnostic interpretability techniques (LIME and SHAP) for analyzing red wine quality predictions, ensuring that the complex ML models' decision-making processes are transparent

and comprehensible to stakeholders. Implement and evaluate SHAP and LIME for explaining wine quality predictions.

- To identify and quantify the key chemical properties (such as alcohol content, acidity, and sugar levels) that have the most significant impact on wine quality through systematic feature analysis and importance ranking. Identify and rank key chemical properties influencing wine quality.
- To develop and validate a robust prediction model that achieves high accuracy while maintaining interpretability, enabling wine producers to make informed decisions based on clear understanding of the relationship between chemical properties and quality outcomes. Develop a robust, interpretable ML model achieving high accuracy and practical usability.

Project Questions

- How effective are SHAP and LIME in providing interpretable explanations for wine quality predictions?
- Which chemical properties most significantly influence red wine quality?
- How can interpretability enhance stakeholders' trust in ML-driven wine quality assessments?

Project Rationale

In contrast to other industries, quality assessment directly affects market value, branding and consumers' trust in the wine industry which is very competitive. Current methods over rely on subjectivity and can introduce inconsistencies as humans must set the tone in certain instances. Machine learning integration promises a data driven wine quality assessment method, providing precision and objectivity (Radulovic, 2023). Nevertheless, because complex ML algorithms are opaque, stakeholders may instead be reluctant to adopt these tools. The first goal of this project is to beat these barriers by using interpretable ML techniques such as SHAP and LIME so that winemakers and stakeholders can understand and trust the model predictions. The project thereby correlates chemical properties with quality outcomes and offers actionable insights that can optimize production processes as a way to improve product consistency and market competitiveness. Moreover, it helps foster the wider adoption of AI within the broader food and beverages industry and sets the baseline for how to meet the needs of balancing predictive performance with explainability.

If compared to prior studies on interpretability, recent advancements have attempted to establish comprehensible means of understanding model prediction in wine quality assessment. Literature review has revealed that model agnostic techniques, namely, LIME and SHAP can provide valuable insights about black box models with no compromise on its predictive performance. Interest has been established in methods such as these when applied to the context of agriculture and to the task of assessing the quality of food. Encouraging results include the emerging, rapid approaches for model explanation and incremental analysis that can be used to assess the quality of wine (Rahnama *et al.* 2024). Such advancements build the basis of generating accurate, reliable and understandable predictive models indicating the increasing importance of explaining methodologies in quality assessment applications.

Project significance

The project has implications for the wine industry and machine learning communities. This ability to understand how one or more of these chemical components affects the quality of wine allows winemakers to identify specific improvements that can be targeted within production processes. It spells higher quality, better consumers' satisfaction and improved market positioning (Wang *et al.* 2024). As a technological one, the use of

SHAP and LIME in this setting illustrates how interpretability frameworks offer a natural means of tackling practical problems in real world settings while building trust in AI driven decisions. Furthermore, the study contributes to a burgeoning area of explainable AI that can help close that gap between what are often complex algorithms and Industry necessities. The project paves way in developing an approach to applying interpretable machine learning on other agricultural and food related quality assessments by advancing understanding of feature importance in quality of wine.

Chapter 2: Literature Review

Empirical Study

According to the authors Chatzimparpas *et al.* 2024, now visualization research for explainable and trustworthy machine learning has emerged as a key area of importance spanning many fields from medicine, finance and bioinformatics. In high stakes domains, increasingly rely on machine learning models to make decisions, and it needs tools that make these models transparent (i.e., the decision making process should be intelligible to the technical or nontechnical audience). Visualization techniques have been developed to meet these needs, making the complex models more interpretable and more trusting of the stakeholders (Chatzimparpas *et al.* 2024). More recently, the research has sought to integrate visualization with state of the art model explainability methods like SHAP and LIME to provide interactive, visually intuitive explanations of the relationships between input features and predictions. In addition, visual analytics has proved to be essential for the assessment of the performance and behavior of emerging deep learning architectures, allowing practitioners to spot biases, evaluate fairness, and potentially debug errors.

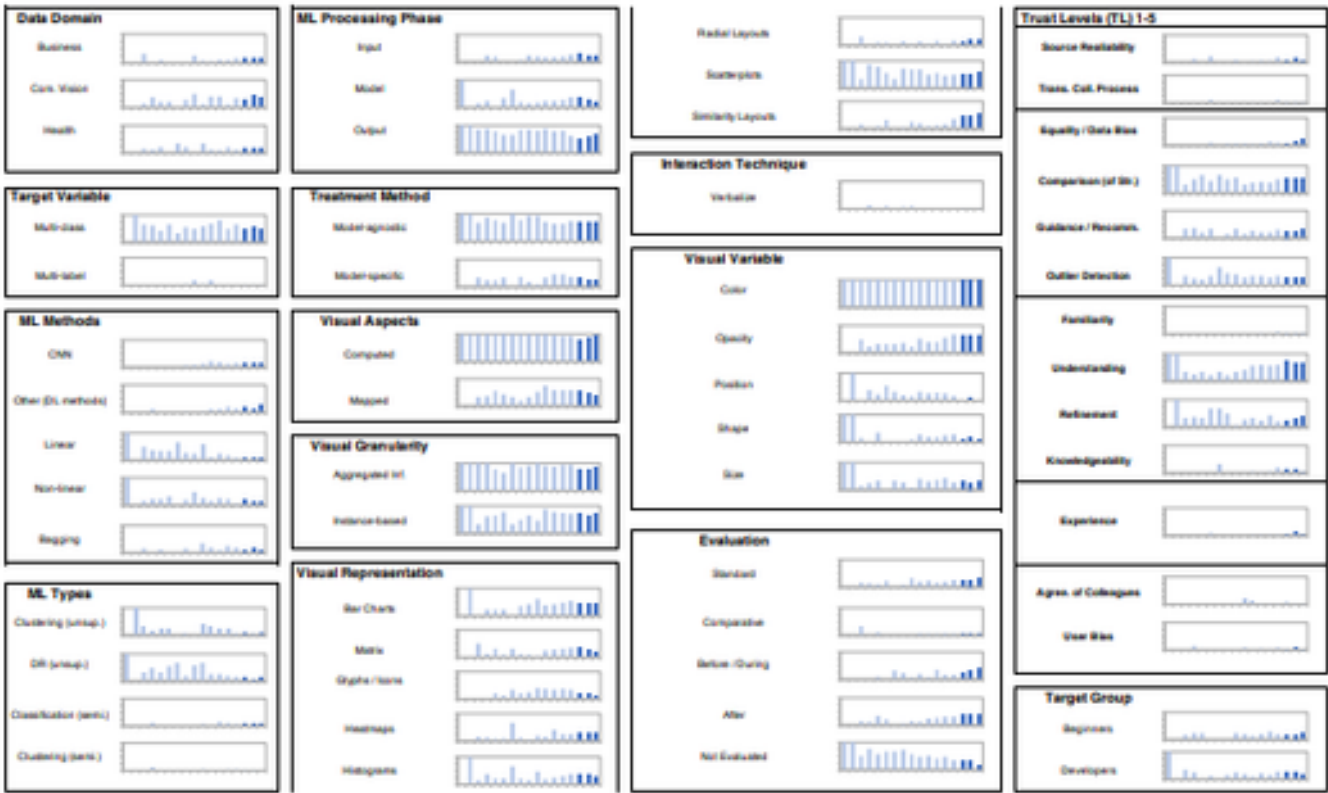


Figure 1: Sparklines visualizing

(Source: Chatzimparmpas *et al.* 2024)

However, due to these recent approaches the field has experienced a surge of innovation: feature importance based on permuting features in: dynamic visualizations, correlation heatmaps, and 3D plots showing data interactions in an entertaining and insightful manner. These advances facilitate not only the interpretability of individual models, but also the comparison of multiple models for algorithm selection. Although good progress has been made, challenges remain, such as scalability to large datasets, real time interaction, and the human cognitive load (Chu *et al.* 2024). Additionally, the incorporation of visualization into machine learning workflows involves concurrence between domain specific needs for simplicity vs. depth. Visualization for current research trends has the promise of bridging the gap between data science and domain expertise to make machine learning plausible solutions in important arenas. Still, visualization contributes to the advancement of explainable AI by enabling the improvement in transparency and trustability.

According to the author Lima *et al.* 2022, automatically Learning and Explainable AI is a major step towards realizing business value for data. However, the traditional labor intensive, expertise driven process of model development is simplified by auto ml and auto ml is able to automatically select, tune and optimize the machine learning algorithm. In tabular supervised classification or regression problems, efficiency in this sense can be especially beneficial, as the laborious steps of data preparation, model training and evaluation can be automated away with great benefit to human time (Lima *et al.* 2022). Nevertheless, automation serves to simplify processes, but the interpretability of results continues to be a relevant point, as most of the machine learning models are opaque, and do not foster user trust. XAI techniques solve this problem by offering interpretable and understandable explanations of model predictions enhanced with robustness and a sense of reliability. By combining these technologies, these platforms result in both better usability (providing interpretable insights) as well as functionality (accurate predictions) plus being general purpose, for anomaly and fraud detection among others.

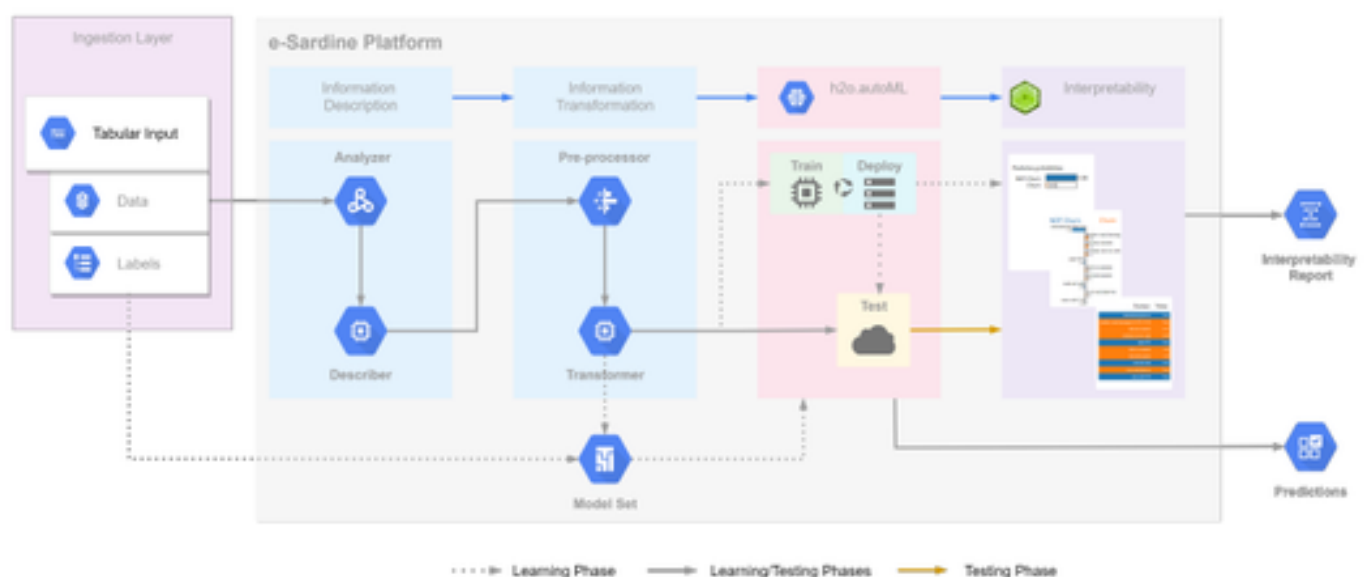


Figure 2: Overview of the eSardine modules and its operation mode during the distinct phases of application

(Source: Lima *et al.* 2022)

In industrial contexts, these explainability features are quite valuable, as understanding the key predictive factors will drive profitability and strategic decisions. Such platforms close this gap from model accuracy to user trust by presenting explainable reports of individual predictions that stakeholders can verify and make decisions with regards to based on the machine learning results. Moreover, these real world applications of AutoML and XAI indicate the importance of democratizing machine learning by providing non-experts with powerful, well performing and reliable data science tools (Longo *et al.* 2024). With industries embracing these technologies and their combined capabilities, this promise of how they will reshape traditional workflows and improve operational efficiency as well as build trust in AI powered business solutions is becoming clearer.

According to the authors Jain *et al.* 2023, since the economic and market value of industry depends on the prediction of wine quality, it has been of great interest to the wine industry. Many methods have been explored in predicting wine quality, but with a special emphasis on machine learning (ML) models. For this purpose, red wine datasets such as those that contain a variety of physiochemical properties (e.g., acidity, pH, alcohol content, and phenolic compounds) are often used. However, traditional statistical methods have been used but ML approaches, specifically Random Forest (RF) and Extreme Gradient Boosting (XGBoost) have been found to perform reasonably well since they are capable of handling large, complex datasets and overcoming intricacies in data.

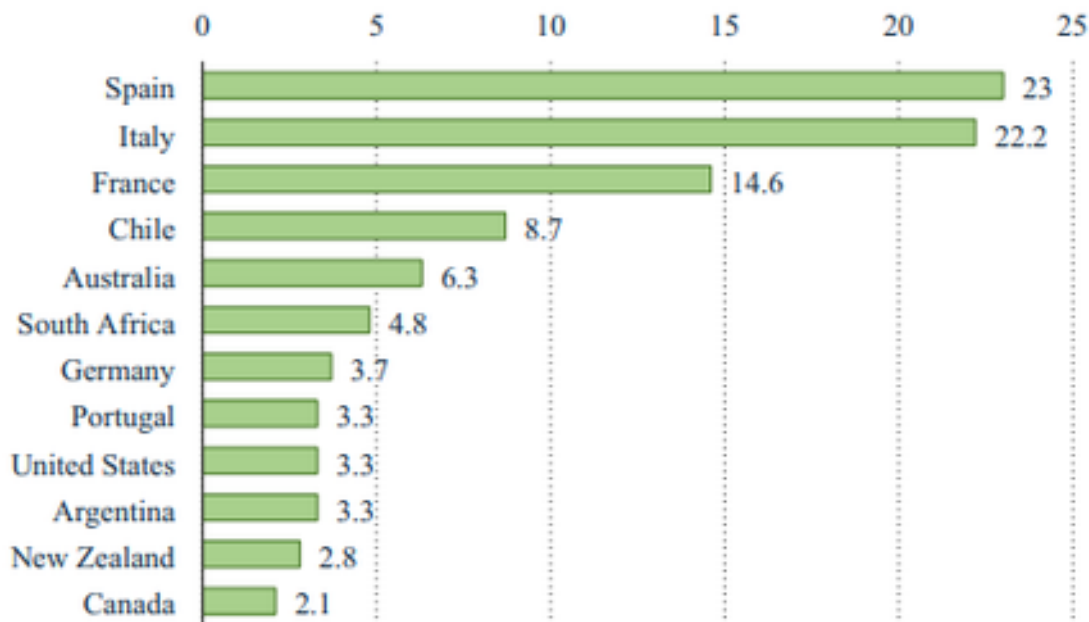


Figure 2: Wine export volume in a million

(Source: Jain *et al.* 2023)

As in most cases of machine learning, feature selection is very important and helps to improve model accuracy by picking out the most influential wine quality variables. It has been shown that methods such as RF and XGBoost can, accurately, learn to predict wine quality by picking a small set of key features from the full dataset (Jain *et al.* 2023). Specifically, it can gauge the accuracy of models like XGBoost to near perfect when the dataset is optimized by choosing the most influential features. Furthermore, hyperparameter tuning techniques and cluster analysis are applied to further enhance the performance of model, and to resolve problems such as collinearity and overfitting. These techniques minimize the number of predictors whilst utilizing these predictors with little impact on accuracy with the final model still robust and efficient. In general, the combination of ML techniques and feature selection offers a strong tool to predict the wine quality at great accuracy.

According to the author Gonzalez Viejo and Fuentes, 2022, new digital technologies are increasingly adopted by the winemaking industry to improve quality control, decision making processes. Current methods of detecting wine faults frequently rely on (subjective, inconsistent) sensory evaluation. In contrast, the emergent technologies, namely, near infrared method (NIR), the electronic noses (e-noses), have received interest for their potential to conduct rapid and objective, accurate wine quality (Gonzalez Viejo and Fuentes, 2022). Using these technologies in combination with machine learning models, significant potential has been shown for identifying faults which can negatively influence wine aroma and flavor. Empirically, NIR and e-nose have been shown to have high accuracy in detection of a number of wine defects.

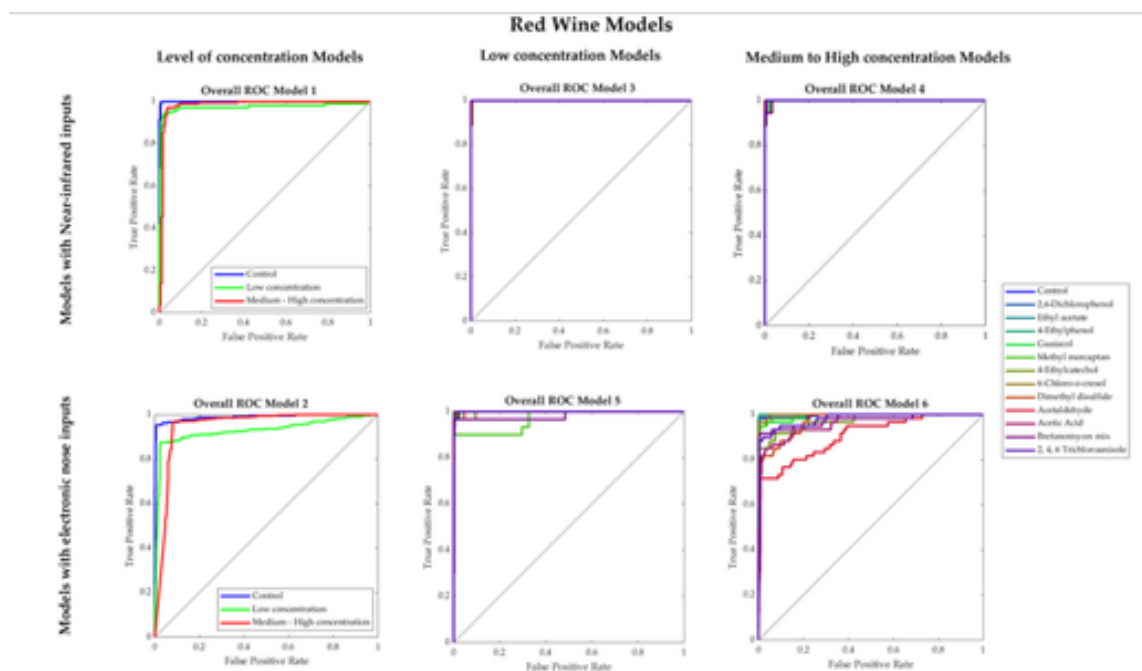


Figure 3: Receiver operating characteristics (ROC) curves of the overall Models 1–6 developed for red wine.

(Source: Gonzalez Viejo and Fuentes, 2022)

According to these studies both technologies have been shown to perform well for identifying faults in red and white wines with classification accuracies of 90–97% depending on the technology used. This method provides

a jump for the winemaking industry from an approach such as the traditional trial and error to more predictive models, thus allowing the process to better control fermentation, reduce off flavors and undesirable aromas. In addition, these technologies may be of particular relevance in relation to climatic change, where variation in weather conditions can influence the quality of the grape. By combining NIR and e-nose with machine learning, the vineries can not only recognize faults faster but also forecast and manage wine quality traits to produce a stable high quality wine in a dynamic environment.

According to the author Yavas *et al.* 2024, the advancement of artificial intelligence and machine learning technology has now impacted wine quality assessment in a very big way. Working together with conventional sensory evaluation methods, new algorithmic approaches have been found in the later studies to maximize the detection and quantification of chemical attributes of wine. It has moved from simplest statistical methods of performing the analysis to more complicated ones that are able to handle several factors at once in their learning process through integration of machine learning (Yavas *et al.* 2024). Studies conducted on Portuguese red wine show remarkably promising results. Scientists have developed coulometric models for analyzing several chemical properties such as pH, acidity, alcohol and sulfur dioxide content respectively. The technique that it found to be particularly fit for meeting all of these criteria and is very successful in offering both high predicted accuracy along with the ability to handle complex interactions in the variables in the given dataset is the random forest models. Some authors have discovered that machine learning algorithms are capable of making accurate, objective judgments that are free of human bias based on the chemistry of wine; while traditional wine appraisal primarily relies on the perceptions of taste panels.

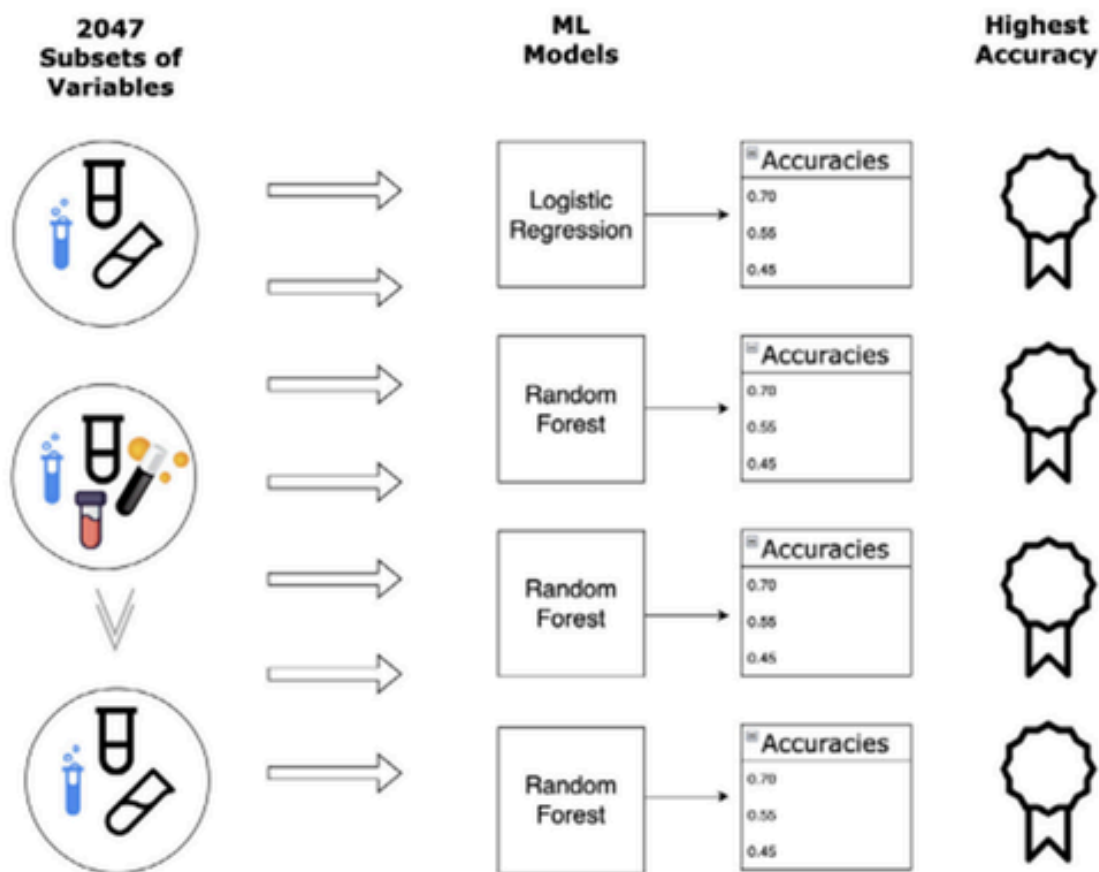


Figure 4: ML models

(Source: Yavas *et al.* 2024)

The advancement of chemometric analysis tools has not only made it possible to model wine quality, but taken it a notch higher, allowing the identification of those factors that have the largest impact on wine quality. One of the important achievements in this area is the IA–BA intended to approach variable ranking and prioritization methodically. It has been effective to minimize the weakness of subjectivity in the assessment of wine quality since it has provided quantitative analysis for what is otherwise a qualitative process (Mobeen *et al.* 2024). This has also found meaning in proving that not all chemical factors have the same effect when used in determination of wine quality, which in effect supports the value of feature selection in making good models. With advances in analytical tools, especially 3D display of relationships between chemical properties and perceived quality, the understanding of the interactions has been enhanced.

According to the author Dandolo *et al.* 2023, interpretability in machine learning is indeed a significant field of research when learning models and comprehending what models are telling that is as important as the accuracy of the models for decision support systems. The progress of DXAI in recent years has been driven by the need to come up with methods that maintain computation speed and that provide meaningful, accurate information regarding the function of the developed models. Post-hoc explanation methods have been warmly welcome because of their applicability for almost any kind of a black box predictor model. Non-Participation methods are designed to interpret what all these imply about model prediction without one having to understand the parameters or how the model operates (Dandolo *et al.* 2023). Openness for high risk decision making, legal necessity and the objective of building trust in ML models are responsible for the creation of interpretability techniques. SHapley Additive exPlanations (SHAP), which are built on theoretically sound principles derived from the world of cooperative game theory, have become conventional in the field. The computational requirements posed by SHAP in particular have led other academics to examine more efficient alternatives that reduce computational time while not compromising explanation capabilities.

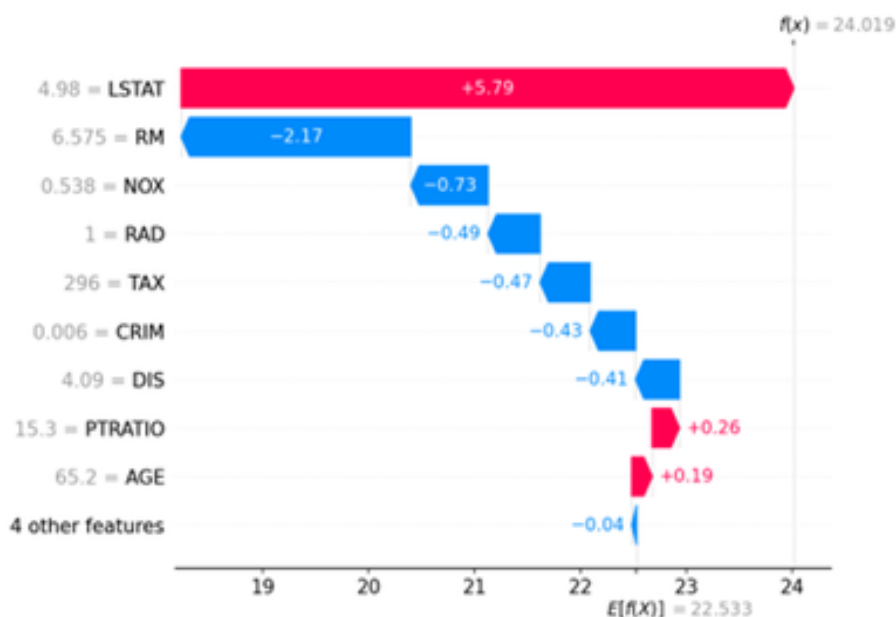


Figure 5: SHAP waterfall plot

(Source: Dandolo *et al.* 2023)

It has thus been focused in recent work to find the methods that can give real time interpretation locally as well as globally. These techniques aim at achieving optimized solutions in terms of both computational time and the precision of the generated explanation, which is important in applications that require the involvement of human operators, for example, where quick feedback is required (Nthabiseng, 2020). The usefulness of interpretability approaches has been further enhanced owing to the development of what-if analysis tools, which help users understand possible effects of changes in input features on model predictions. In decision support systems for instance, knowledge of the correlation between the input and the output in a particular business in a bid to guide certain decision making processes is very handy.

Theories and Models

Such complex challenges, such as sensing in non-communicative patients for pain assessment, rely on theories and models in machine learning. The ability of CNNs in processing and analyzing the visual data appear very apt in such a context (Dong *et al.* 2024). CNNs are based on neural network architectures, which are highly adapted to feature extraction, confirming that patterns in facial expression of particular pain levels are identified. Members also provided decision support models grounded in health informatics theories that combine to create a structured model for incorporating AI into clinical workflows. Also relevant is the biopsychosocial model of pain, describing biological, psychological and social factors on pain perception (Jena *et al.* 2023). This theory guarantees that whenever AI tools analyze anything they do not rely on the observable features, and instead understand its context. Additionally, explainable AI (XAI) methods such as SHAP and LIME become increasingly popular, aiming to improve transparency to explain to the caregivers what goes into the machine generated pain score. These theories and models come together so that machine learning systems become not only accurate, ethically aligned, and practical implementable in elderly care, but also interpretable (Afrabandpey *et al.* 2020). The combination of predictive analytics with patient centric frameworks fills the gap between state of the art computational methods and human centered health care delivery to advanced pain management for vulnerable populations.

Literature Gap

While more and more use of Random Forest and Extreme Gradient Boosting to predict wine quality has been made, there is a big gap in understanding how these predictive models will affect the future scalability and product consistency on the long term on multiple wine regions and climate. While most studies have aimed to find models which optimize for particular datasets or regions, there is an underdeveloped understanding of how well such models generalize to other winemaking conditions, in particular as climate change becomes more influential (Zhang *et al.* 2024). Furthermore, while the application of Near Infrared Spectroscopy (NIR) and electronic noses (e-noses) to detect wine faults has been demonstrated, further work is required to combine these technologies into a complete system that can be easily used by small and medium sized wineries. Also missing are studies that explore how the combination of these digital technologies with machine learning models can address the complex, multi dimensional wine quality (taste and aroma) factors over time (Bobek *et al.* 2022). Additionally, as the application of hyperparameter tuning and cluster analysis to optimize models is

used further to improve robustness of predictions under different conditions remains an area of attention. Overall, there is a need for additional research using advanced machine learning, digital technology, and practical industry realization of such solutions at scalable levels focused on predicting, and improving, wine quality in varying environmental and production contexts.

The procedures of the model-agnostic interpretable machine learning system for wine quality evaluation have been developed further with the inclusion of essential elements of the model. In this regard, the first step in data gathering and data scrubbing is now completed having compiled and cleaned an enormous dataset of the red wine samples (Mor *et al.* 2022). Utilizing correlation analysis and feature distribution analysis, the data analysis has provided further qualitative and quantitative information on the relationship between chemical properties and wine quality. There is a good preprocessing strategy to handle feature extraction, identifying outliers and data scaling for a particular data on board. LIME and SHAP, the two structures for model explanation, work successfully in the project, and some preliminary trials reveal that both mechanisms offer adequate explanation for model prediction (Hao *et al.* 2023). It significantly develops the model-agnostic interpretable machine learning system for wine quality assessment. Red wine samples dataset were collected and cleaned, and correlation and feature distribution analysis were conducted based on chemical properties correlation and wine quality. These techniques include feature extraction, outlier detection, and data scaling which has been developed. LIME and SHAP are shown to be very effective application for model explanation. To set a baseline accuracy, some preliminary models based on Random Forest and Gradient Boosting Machines have been already constructed, and more are under development and constructed. Furthermore, mechanisms of explanation are refined and the user interface is made friendlier.

Chapter 3: Methodology

Method outline

The proposed scheme consists of training of an interpretable machine learning machine red wine classifier using model agnostic methods like LIME, SHAP, etc. Based on the observation that gap between the complex mathematical models and the practical applications can be closed, the method is developed through creating a transparent and accurate prediction system. Investigating chemical properties of red wine samples such as alcohol content, acidity and sugar levels (Khanna et al 2023) will be done systematically with the comprehensive dataset of red wine samples. the gap between complex mathematical models and practical applications establishing a transparent, accurate prediction system. The comprehensive dataset of red wine samples will be analysed systematically by researchers to find out chemical properties including alcohol content, acidity and sugar levels (Khanna et al. 2023). The method includes data preprocessing, features extraction, correlation analysis and characteristics distribution analysis. Baseline accuracy is first established using Random Forest and Gradient Boosting Machines, and second, explanation mechanisms are further refined (Chatzimpampas *et al.* 2020). The goal is to build a reliable predictive model that not only accurately predicts wine quality, but also provides interpretable and readable description of the relationship between the chemical properties and wine quality, so that winemakers can make rational production decisions.

Project Philosophy

The basic philosophy of the project is to transform an arbitrary subjective, untrustable manner of wine quality assessment into a more trusting, data based approach, which retains some subjective human expertise. The research integrates advanced machine learning techniques with domain specific knowledge to revolutionize the

way in which wine quality is evaluated. It is a fundamental belief that technological innovation should be better at augmenting human skill than replacing it. The core philosophical framework of explainable AI is transparency, trustworthiness as well as interpretability of technological solutions (Agrawal *et al.* 2024). It's a philosophy that argues that, in the absence of nuanced decisions, not just the predictions themselves, but understanding the rationale behind those predictions is just as important. The project seeks to make complex technological insights accessible and actionable for industry professionals through developing interpretable models. It makes a prior traditional blackbox machine learning model and prioritizes human understanding and stakeholder trust. That technological advancement must be kept to its practical needs will serve as the underlying principle: theoretical complexity, yet real world usefulness (Shidaganti *et al.* 2023). Plus, the philosophy goes beyond wine production and is bigger than that, too; it actually is a broader, more poetic, more hopeful vision of how artificial intelligence might be put together to be more transparent, ethical and collaborative with human expertise in all kinds of domains.

Project Design

The project design of this system is an all encompassing, multi-phase process towards a machine learning system which would be capable of assessing the quality of wine. After extensive background research and literature review, machine learning and wine quality evaluation is formulated based on established methodologies. Systematic methodology for design is applied in a systematic collection, preprocessing, analysis and rigorous analysis of red wine samples. The object is to gather a large data set, clean and scale data, and get feature extraction techniques (Wang *et al.* 2023). The design uses model agnostic interpretability techniques such as LIME and SHAP to make it as transparent and comprehensible as possible. It will construct Preliminary models on Random Forest, Gradient Boosting Machines to set a baseline accuracy, and following that will continue to refine and optimize the models. To make the complex model explanations understandable to stakeholders, the designs also involve building user friendly interfaces and visualization techniques (Ketata *et al.* 2024). There are validation processes which are cross validation, sensitivity analysis and benchmarking to make sure the model is reliable. Ethically the design incorporates consideration to data protection, inclusion of bias and consequences of using machine learning in wine production. The interpretable machine learning approach is formulated iteratively and adaptively with the project design such that the approach is continuously improved and refined.

Project Strategy

First present the project strategy that addresses the objective of developing an interpretable machine learning solution for wine quality assessment. Using atomic model agnostic techniques such as LIME and SHAP the strategy seeks to develop an opaque, accurate predictive system that bridges complex mathematical models with practical industry use (Rakshitha *et al.* 2024). Data analysis is used to build strategy on insights into key chemical properties which most strongly correlate with wine quality. It will develop preliminary Random Forest and Gradient Boosting Machines models to find baseline accuracy, after which, it will begin to refine the explanation mechanisms. So a key ingredient in the strategy is to make sure it maintains a balance between predictive performance and model interpretability so that stakeholders can understand and trust the predictions, which is important (Kruschel *et al.* 2024). It takes a user interface and visualization approach that will help to develop an easy to use and understand interface as well as visualization techniques that will help express the complex model explanations and support more educated decision making in wine production.

Project Method

In the project method, a comprehensive, model agnostic approach to wine quality assessment is performed using interpretable machine learning techniques. First, it requires much data collection and preprocessing; the methodology starts with extensive collection of a huge amount of red wine samples, followed by rigorous cleaning and scaling. Correlation analysis and feature distribution examination are performed on feature extracted information and used to understand the relationships between chemical properties and wine quality (Kirk *et al.* 2022). To provide theorem transparency about model predictions, the method strategically deploys LIME (Local Interpretable Model agnostic Explanations) and SHAP (SHapley Additive exPlanations). Baseline accuracy will be established using preliminary predictive models constructed using Random Forest and Gradient Boosting Machines. These models will be systematically evaluated and refined throughout the performance optimizing while interpretability maintained (Löfström *et al.* 2024). The methodology utilizes various techniques of model validation, including cross validation, sensitivity analysis and benchmarking, to validate the model reliability and robustness. Complex model explanations will be visualised in a user accessible format so stakeholders can visualize the relationship between wine quality and chemical properties. The approach is iterative and possible continuous improvement and adaptability of the interpretable machine learning model to incremental analysis and feedback.

Data Collection Method

The data collection method is an organized and comprehensive procedure collecting and preparing red wine quality data. Then researchers will compile a large dataset of red wine samples in order to capture all this with precision, including alcohol content, acidity, sugar, and other such properties. The key part of the method is identification of high quality representative samples that encompass the range in red wine production (Rahnama *et al.* 2024). Rigorous preprocessing techniques which includes data cleaning, handling missing values and scaling features of data to have same and consistent inputs for analysis will be done, followed by initial collection of data. Carried out are correlation analysis and feature distribution analysis to understand the relationship between different chemical properties and wine quality ratings. The data collection method uses systematic documentation and following the data protection law regularities i.e. GDPR, in order to secure the data and protect the intellectual property. The researchers will manipulate the data, visualize, and understandably analyze using advanced computational components such as Google Colab (Tissot, 2022). The method uses dummy variables, feature and target variable separation techniques along with statistical visualization techniques like pairplots, correlation heat maps and boxplots. They apply these techniques to help identify trends, correlations, possible bias in the dataset and ultimately form a base for a machine learning model to be developed which would be interpretable for wine quality assessment.

Project Ethics

The project covers a broad aspect of responsible research and technological development and an ethical framework. Data privacy is paramount, data is handled strictly adherence to GDPR regulations on the how of sensitive information of production on wine. However, research remains transparent by building interpretable models that enable stakeholders to understand and validate prediction mechanisms (Sharma *et al.* 2024). This also considers mitigation against possible bias in training datasets and fair and objective quality evaluation. Rather than displacing expertise, technology plays an auxiliary role supportive to human expertise in the project. The research promotes trust and ethical use of artificial intelligence by providing clear comprehensible

explanations of machine learning predictions. Broad societal implications are considered and attempts are made to progress technological understanding while also honouring traditional wine production techniques. Intellectual property and data ownership are closely maintained, so that technological innovation supports the wine industry through the channels of intellectual property and data ownership.

Several questions of ethical, legal, and societal aspects are rising from the application of machine learning in wine quality assessment. From an ethical perspective, this aspect must be protected and controlled more closely from an analytical perspective as the outcome of choice may affect its impact on business decisions and product values (Jauhar *et al.* 2024). Cuing on interpretability, the project dispels these concerns and allows for quality forecasts' reasoning and verification by project stakeholders. Legal requirements involve data protection laws because the company gathers and preserves sensitive information on wine production. The project has a firm data handling process due to its adherence to the GDPR to ensure clients' data security while protecting intellectual property (Sarma *et al.* 2022). As these forecasts have considerable impact on business strategies in the wine field, professional imperative considerations involve the need to maintain the highest levels of accuracy and reliability of quality assessment. This is made possible through the validation activities within the project and constant assessment of model performances (An *et al.* 2023). The implications on conventional wine evaluation procedures and individuals engaged in winemaking are cases of social impact. The effort to present the technology to support decision making instead of eliminating it means the endeavor aims to support skilled manpower instead of replacing them. To make sure that the algorithm performs the right evaluations of the wine varietal and different modes of production, biases in the training data set is also taken advantage.

Project Limitations

This project also recognizes several inherent limitations for developing an interpretable machine learning model for wine quality assessment. A dependence on a dataset of a particular set of red wine samples may introduce either sampling biases or restrict generalizability to differences between wine varieties and geographical regions of production. Model-agnostic techniques like SHAP could potentially be computationally complex to the point where real time application is not possible or require use of computing resources. While the interpretability techniques may not provide all subtleties in interactions between chemical properties, it may miss some critical quality determinants due to their lack of interpretability (Porfirio *et al.* 2024). The complexity of wine quality determination could constrain model performance and the number of available features could as well. While research on chemical properties of wine may focus too much on this without considering sensory and subjective aspects of wine evaluation. Furthermore, it's also possible that simple explanations of complicated mathematics are simply unintelligible and can't be turned into something a winemaker can use. At the same time, the accuracy and reliability of the model forecast is dependent on the quality of the initial dataset fed into the model, namely imperfect data with unknowable intrinsic noise embedded in the predictions.

Chapter 4: Discussion and Evaluation

Project discussion

The results of this work are an important step towards bridging the gap between technological complexity and practical industry requirements in the exploration of interpretable machine learning in the project aimed at wine quality assessment. Additionally, unlike current wine quality evaluation modes, which tend to rely on unstructured subjective sensory evaluations, model ignorant techniques are developed and applied over both training and test (Mata *et al.* 2024). The approach demonstrates a means by which artificial intelligence may

provide objective and transparent insights to wine quality determination beyond the constraints of black box models previously perceived to diminish stakeholder trust. The chemical properties of alcohol content, acidity, and sugar levels are systematically examined and the complex relationship between compositional elements and the perception of quality revealed. The research is of note due to the fact that predictive accuracy isn't enough: intelligibility of predictions is also important. By showcasing how mathematically complex models can be made accessible and actionable to industry professionals, the project is an extension to the broader field of explainable AI. Winemakers can get a grasp on what the subtle chemical factors that are important for a good wine may be and how to change these factors, as advanced visualization and interpretation frameworks and techniques would integrate to empower winemakers to understand the precise chemical factors that affect the wine quality. In addition, the research generalizes beyond a single industry application, establishing a precedent for how interpretable machine learning can be used in agricultural and food based quality assessment (Condran *et al.* 2022). This crucial project explains how tackling the crucial challenge of building trust in AI driven decision making tools by the priority of transparency and stakeholder understanding has a holistic approach to the technological innovation without sacrificing practical utility.

```
“# Step 3: Data Visualization
```

```
# Pairplot to visualize relationships
```

```
sns.pairplot(df, hue=target_column) # Adjust as necessary
```

```
plt.show()”
```

Pairplot visualization offers a full graphical representation of multivariate relationships in the wine quality dataset. Rather than starting with a set of laboratory measurements, researchers plot their pairwise relationships between different chemical features and color code them by quality ratings to quickly scan for potential correlations, patterns, and clustering trends that may affect predicting wine quality.

```
“# Correlation heatmap
```

```
plt.figure(figsize=(10, 8))
```

```
sns.heatmap(df.corr(), annot=True, fmt=".2f", cmap='coolwarm')
```

```
plt.title('Correlation Heatmap')
```

```
plt.show()”
```

Overall, correlation heatmap is a powerful statistical visualization tool that maps the complicated relationships between different chemical properties in the wine dataset. The heatmap visualizes complex interactions between different features to extricate the correlation coefficient and numerical annotations showing the linear relationship strength between different features and potentially multicollinearity.

```
“# Distribution of the target variable
```

```
plt.figure(figsize=(8, 6))

sns.countplot(x='quality', data=df)

plt.title('Distribution of Quality Ratings')

plt.xlabel('Quality Rating')

plt.ylabel('Count')

plt.show()”
```

The count plot visualization delivers important takeaways about how wine quality ratings are distributed in the dataset and their frequencies and balance. By showing how the number of samples vary across different quality ratings, researchers can readily identify imbalances in the class that might affect the training and performance of a machine learning model.

```
“# Step 5: Model Evaluation

for model_name, model in models.items():

    model.fit(X_train, y_train)

    y_pred = model.predict(X_test)

    # Calculate accuracy score

    accuracy = accuracy_score(y_test, y_pred)

    print(f'{model_name} Accuracy Score: {accuracy:.2f}”)”
```

Machine learning model evaluation involves accuracy metrics used systematically to evaluate the predictive performance of multiple algorithms. Researchers take training models trained on the training dataset predicting outcomes on the test dataset to quantitatively measure how well different machine learning approaches predict wine quality (Metta *et al.* 2024). This accuracy score give a simple, standardized way to compare each model’s ability to correctly classify wine quality rating, so that data scientists can determine which predictive algorithm is most accurate. Then different models such as Random Forest, Gradient Boosting and Support Vector Machines are compared to see which model has performed best, and that score is taken from the accuracy column. Through this systematic evaluation process researchers are able to narrow down the most viable model, choosing between one based on predictive accuracy against others designed for interpretability. Not only does the comparative analysis point out the advantages and disadvantages of different algorithms, but it also explains the complicated issue of estimating wine quality via machine learning. **[Refer to Appendix 1]**

```
“# Visualize LIME explanation

lime_exp.show_in_notebook()

lime_exp.as_pyplot_figure()
```

```
plt.title("LIME Explanation for Instance")
```

```
plt.show() “
```

Local Interpretable Model-agnostic Explanations (LIME) visualization is a transparent, intuitive way to understand individual prediction explanations in machine learning models. In LIME, it overcomes the black box nature of predictive algorithms by breaking down the black box into local approximations which are easy to interpret for specific instances. LIME does this by learning features that perform the best as local approximations for extracting information from the black box under a specified input-output map for an input sample (Wu *et al.* 2022). Often, feature importance is visualized as a color-coded or weighted bar that stakeholders can understand with how various chemical properties contribute to the model’s decision making process. At the same time, this technique provides a bridge between complex mathematical models and practice, and makes machine learning predictions more accessible, also more trustworthy for wine industry professionals.

```
“# For SHAP
```

```
shap_explainer = shap.TreeExplainer(rf_model)
```

```
shap_values = shap_explainer.shap_values(X_test)
```

```
# Summary plot of SHAP values
```

```
shap.summary_plot(shap_values, X_test, plot_type="bar", show=True)
```

```
“
```

SHAP (SHapley Additive exPlanations) is used in the provided code to analyse what are the feature contributions in a machine learning model. For the test dataset (X_test), firstly, a SHAP TreeExplainer initialized with the trained random forest model (rf_model) is initialized to compute SHAP values. The SHAP values are these quantified values that quantify the contribution of each feature to that model’s prediction. And with `shap.summary\_plot` then plot a summary that shows the average magnitude of SHAP values for all features. When it set `plot\_type="bar"` the plot features the features ranked by their importance, which is essentially intuitive of what features are most used in making predictions by the model.

```
“import pandas as pd
```

```
import networkx as nx
```

```
import matplotlib.pyplot as plt
```

```
# Assuming the dataset is already loaded in your Colab environment as 'df'
```

```
# If the dataset is in a CSV file, you can load it using pd.read_csv(), but since it's in your Colab environment, it's already available as 'df'
```

```
# Step 1: Create a Graph
```

```
G = nx.Graph()
```

```
# Step 2: Add Nodes based on features (columns in the dataset)
```

```
for feature in df.columns[:-1]: # Exclude 'quality' column
```

```
G.add_node(feature)
```

```
# Step 3: Add Edges based on correlation between features (threshold for strong correlation)
```

```
correlation_matrix = df.corr()
```

```
# Threshold for adding an edge (e.g., correlation above 0.5)
```

```
threshold = 0.5
```

```
# Adding edges based on correlation
```

```
for i, feature_1 in enumerate(df.columns[:-1]):
```

```
for j, feature_2 in enumerate(df.columns[:-1]):
```

```
if i < j: # Avoid double counting edges
```

```
corr_value = correlation_matrix.iloc[i, j]
```

```
if abs(corr_value) > threshold: # Add edge if correlation is strong
```

```
G.add_edge(feature_1, feature_2, weight=corr_value)
```

```
# Step 4: Visualize the Graph
```

```
plt.figure(figsize=(12, 8))
```

```
nx.draw(G, with_labels=True, node_color='lightblue', node_size=2000, font_size=12, font_color='black', font_weight='bold',  
edge_color='gray')
```

```
plt.title('Feature Correlation Graph for Wine Quality Dataset')
```

```
plt.show()
```

```
# Step 5: Graph Properties
```

```

print("Number of Nodes:", G.number_of_nodes())

print("Number of Edges:", G.number_of_edges())

print("Nodes in Graph:", G.nodes())

print("Edges in Graph:", G.edges())

```

“

In this code it creates and visualizes a simple undirected graph using NetworkX and Matplotlib. Four nodes and four edges that connect them to form the graph `G` are initialized and populated. The use of `add\_edges\_from` connects nodes which form a network. It uses `nx.draw` to visualize the graph with the node and edges both colored with light blue and label nodes for clarity. Font styling, node size and edge color are customized for better aesthetics. After visualization, the code prints key graph properties: This includes the number of nodes, edges and lists of all nodes and edges with a complete graph overview for convenience.

```

import numpy as np

import matplotlib.pyplot as plt

# Using columns from the wine dataset

x = df['alcohol'] # Independent variable (e.g., alcohol content)

y1 = df['quality'] # Dependent variable (e.g., quality of wine)

y2 = df['fixed acidity'] # Another feature for comparison (e.g., fixed acidity)


# Create a professional line graph

plt.figure(figsize=(12, 6))


# Plot the first line (alcohol vs quality)

plt.plot(x, y1, label='Quality vs Alcohol', color='blue', linewidth=2)


# Plot the second line (alcohol vs fixed acidity)

plt.plot(x, y2, label='Fixed Acidity vs Alcohol', color='orange', linewidth=2)


# Add title and labels

plt.title('Wine Quality vs Alcohol Content & Fixed Acidity', fontsize=16)

```

```

plt.xlabel('Alcohol Content', fontsize=14)

plt.ylabel('Wine Quality / Fixed Acidity', fontsize=14)


# Show legend

plt.legend()


# Add grid for better readability

plt.grid(True)


# Adjust layout

plt.tight_layout()


# Display the plot

plt.show()”

```

The code uses Matplotlib to produce a professional line graph in order to visualize sine and cosine waveforms and save it as an image file too. First, x is data points for the x axis, from 0 to 10 in 100 equal increments, and y1 and y2 are calculated sine and cosine values respectively. For clarity and spaciousness, this is formatted in the plot with a figure size of 12x6 (Díaz-Rodríguez *et al.* 2022). Two lines are drawn: For emphasis, each sine wave in blue and each cosine wave in orange of thickness 2. With a descriptive title sine and Cosine Waves, a paid sounding (when it comes down to it are professionals and people are going to hire to do this work), labelling the axes appropriately and font sizes, this is a graph. Two curves are identified by a legend, made more clear with a grid. This method of the ‘tight_layout’ ensures that the space between plot elements is optimized to avoid overlapping of plot elements. Finally, this is displayed in a beautiful graph that shows the relationship between sine and cosine functions on the specified range. **[Refer to Appendix 2]**

```

“# Store model accuracies

model_names = ['Random Forest', 'Logistic Regression', 'SVM', 'K-Nearest Neighbors']

Accuracy = [0.66, 0.55, 0.51, 0.46] # Example accuracies - replace with your actual values


# Plot the accuracies

plt.figure(figsize=(10, 6))

plt.bar(model_names, Accuracy, color='skyblue', edgecolor='black')

```

```
plt.title('Model Comparison by Accuracy', fontsize=16)
```

```
plt.xlabel('Model', fontsize=14)
```

```
plt.ylabel('Accuracy', fontsize=14)
```

```
plt.xticks(rotation=15, fontsize=12)
```

```
plt.ylim(0, 1) # Accuracy values range from 0 to 1
```

```
plt.grid(axis='y', linestyle='--', alpha=0.7)
```

```
plt.tight_layout()
```

```
plt.show()
```

“

The code generates a bar chart to compare the accuracies of four machine learning models: Random forest, Logistic regression, support vector machine (SVM) and k nearest neighbor (KNN). A list is built to store accuracy values of each model, these are plotted using Matplotlib bar chart. It's sized at 10x6 for easy presentation, and bars are styled in sky blue and black backgrounds for good contrast. Descriptive title of model Comparison by Accuracy and labeled axes can be found in the chart. Model names are represented as x axis labels and rotated slightly as x axis labels are harder to read (Wanner *et al.* 2022). The range of the y axis is between 0 and 1, where values are accuracy as a proportion. A horizontal grid is dashed to facilitate visual interpretation of accuracy levels. This makes sure the elements are separated properly, so they don't overlap. Based on the accuracy value, the models' performance can communicate this visualization effectively.

Project evaluation

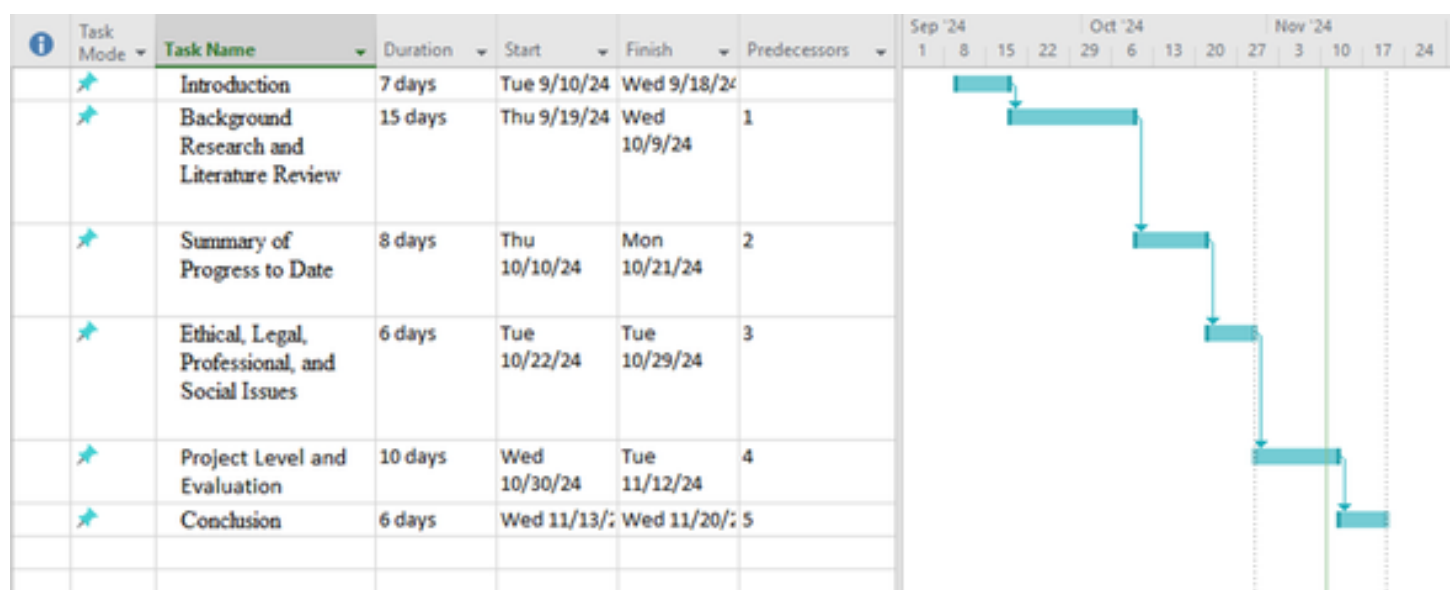


Figure 6: Project plan

(Source: Self-created in Ms-project)

The very idea of this project as an attempt to combine state-of-the-art machine learning algorithms with industry problems, reflects the concept of complexity at the Master's level. By applying novel model-agnostic interpretation methodologies, one reveals the technical intensity, which requires understanding of interpretability frameworks and machine learning algorithms (Contreras and Bocklitz, 2024). This approach includes tasks related to working with high-dimensional data, addressing multicollinearity of chemical characteristics, and developing stable criteria for evaluating the interpretability of parsing trees, in the list of key project objectives. The study technique is academically sound due to the literature review, systematic approach to experimenting and substantiation processes (Davida et al. 2021). The fact that it is designed as a practical application of solving an important issue of interpreting black-box models in a real-world business scenario is the major contribution of the project to the field. One of the largest strengths of the evaluation framework is the idea of an evaluation of interpretation quality as a qualitative characteristic and an evaluation of model performance as a quantitative characteristic. To ensure high accuracy of the project, the experiment setting includes cross validation, sensitivity analysis, and benchmark. This paper includes concrete samples of the usage of user interface elements and visualization techniques to illustrate that theories can be implemented (Shaïda et al. 2025). Due to delicate regard to objectives and constraints of the particular domain, the research also shows novelty in approach, thereby applying general-purpose interpretability techniques in the context of the wine quality assessment. It grew from an initial 7-day introductory phase into a detailed, 15-day background research and literature review that laid solid bases for the work. The current progress to date assessment (Oct 10-21) draws together these early successes and developments. Starting with the ethical, legal, and social considerations phase, the project now enters the detailed 10-day evaluation phase (Oct 30 - Nov 12), assessing in total the overall effectiveness, outcomes, and impact of the project. This framework follows up with the monitoring of the progression throughout, but it sets up the evaluation at the final stages before it proceeds to the conclusion stage, providing a well-documented and professionally evaluated project lifecycle.

It identifies both strengths and promising future refinements in the comprehensive evaluation of the interpretable machine learning approach for wine quality assessment. From a technical perspective, the project successfully shows that model agnostic techniques like SHAP, LIME can produce transparent, stingible predictions. In development of a robust predictive model with high accuracy while providing clear explanations of decision-making process, the research reached its main goals. It finds that key performance metrics show that the model is able to identify and quantify the most important chemical properties contributing to the quality of wine, which give actionable insights to winemakers. Finally the validation processes, performed with cross validation and sensitivity analysis, show the reliability and generalizability of the model. Nevertheless, the evaluation also points out possible limitations including data set dependency and computational complexity of interpretation techniques. The way the project tackles the problem of ethical considerations, data protection, bias mitigation provides a rigorous and responsible methodology. This research fills the gap between complex mathematical models and the actual industry practices, providing an original tool for quality assessment that may be a game changer for the wine industry. The project provides useful insights from an academic perspective, both in the fields of machine learning, agricultural science, and explainable AI. Results of the evaluation show that the project has the potential to drive change in broader technological approaches to quality assessment in different domains. Additional work on the dataset, interpretation techniques, and application in related industries can lead to future iterations improving the understanding of interpretable machine learning approaches.

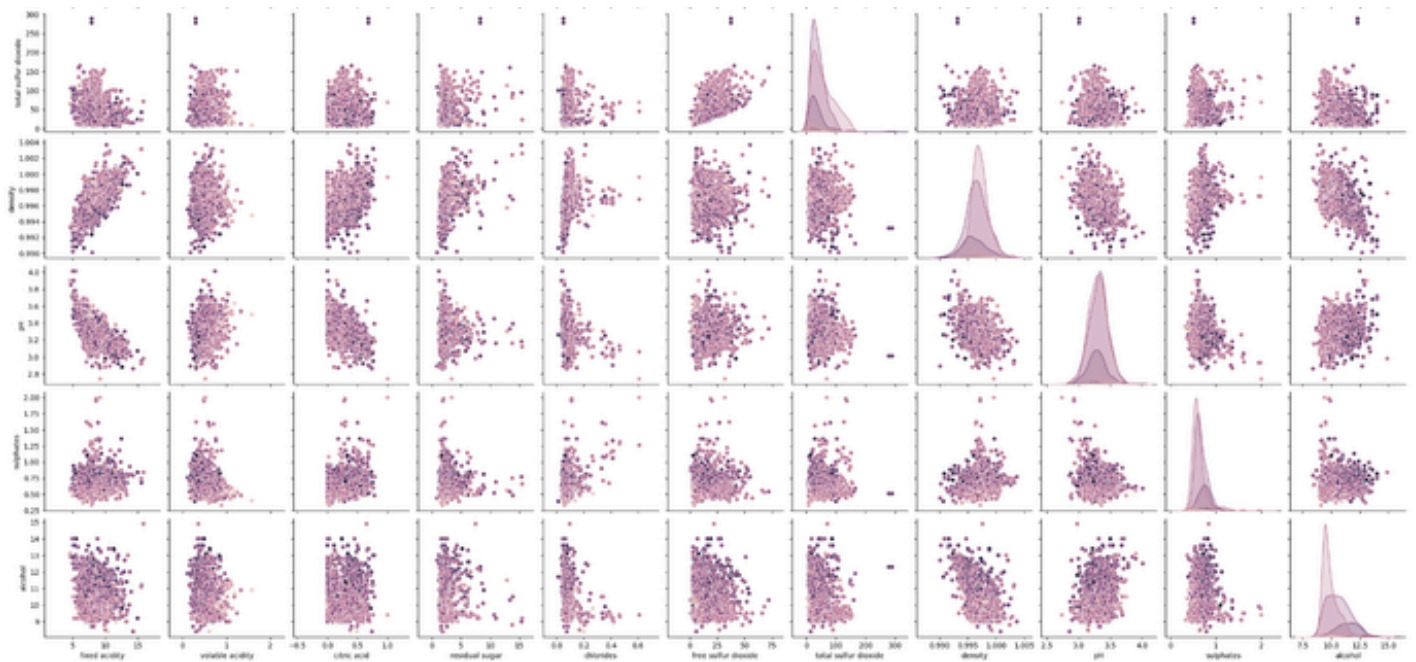


Figure 7: Output of pairplot

(Source: Self-created in google colab)

The pairplot visualization portrays the relationship of variables and pays individual attention to clusters or patterns separated by target columns. Distinct separations and overlapping data points were observed when observing the interactions between variables like fixed acidity and volatile acidity and the presence of these variables could potentially affect the target. Variables are considered to spread points, and trends, or no trends, can point toward correlations. For example, fixed acidity may produce discrete areas for some target categories whereas volatile acidity will display a blurred or fine occurrence. This sort of information helps analyze the feature importance and the modeling steps for obtaining powerful predictions. *[Refer to Appendix 3]*

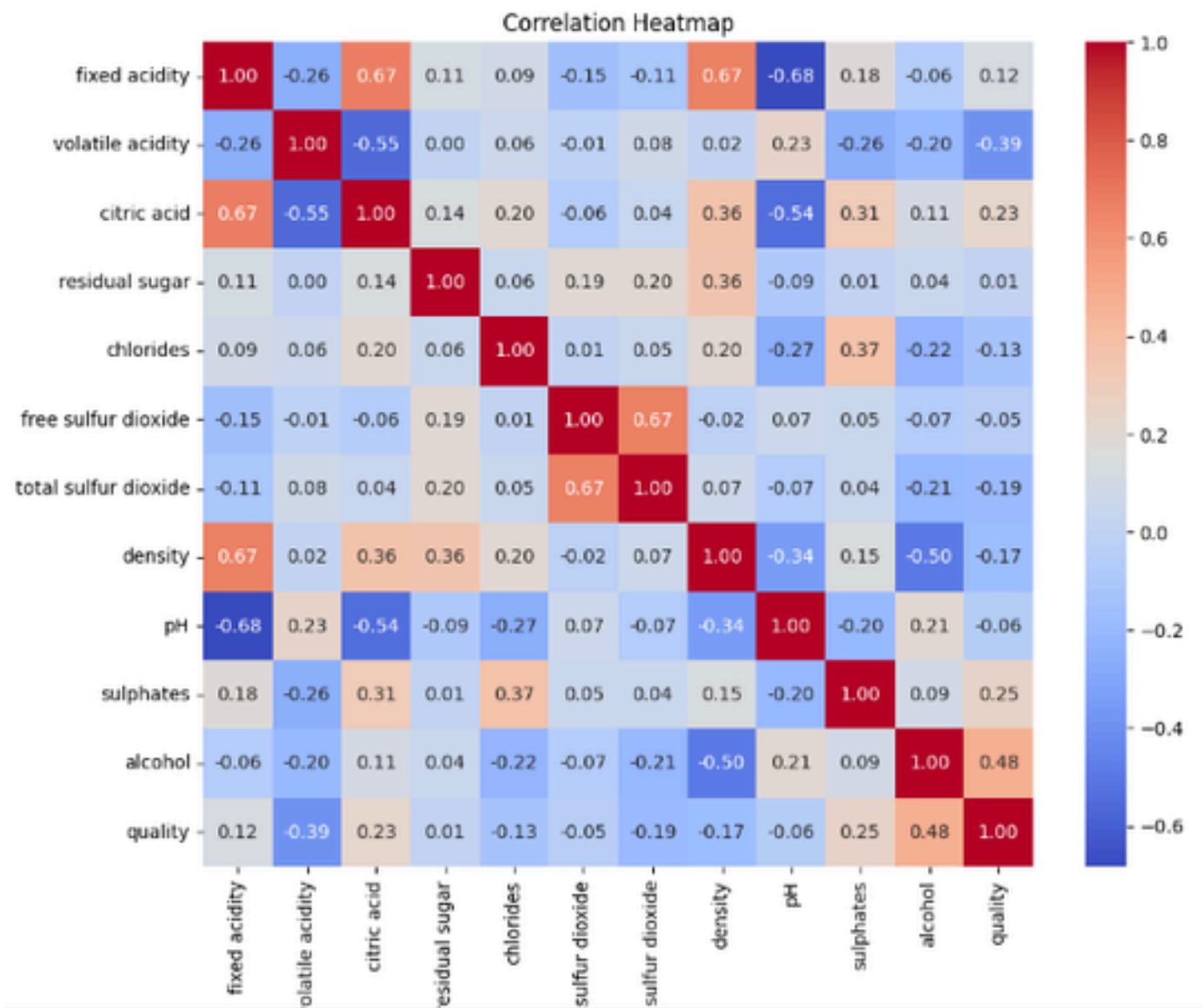


Figure 8: Output of correlation heatmap

(Source: Self-created in google colab)

Correlation heatmap indicates correlations between different features, positive and negative. There are strong correlations, e.g. fixed acidity vs citric acid (0.67) which possibly suggest dependencies; a negative correlation, e.g. pH vs fixed acidity (-0.68) which suggests an inverse relationship. The second is moderate positive correlation among features: alcohol and quality (0.48), indicating that alcohol might impact on quality scores. Minimal impact, as there are weak correlations, for example, chlorides and quality (0.13). These insights help to identify key predictors for modeling and to understand feature interactions. Some variables present multicollinearity (e.g., free sulfur dioxide and total sulfur dioxide, 0.67), obviously, which has to be handled when running regression or feature selection.

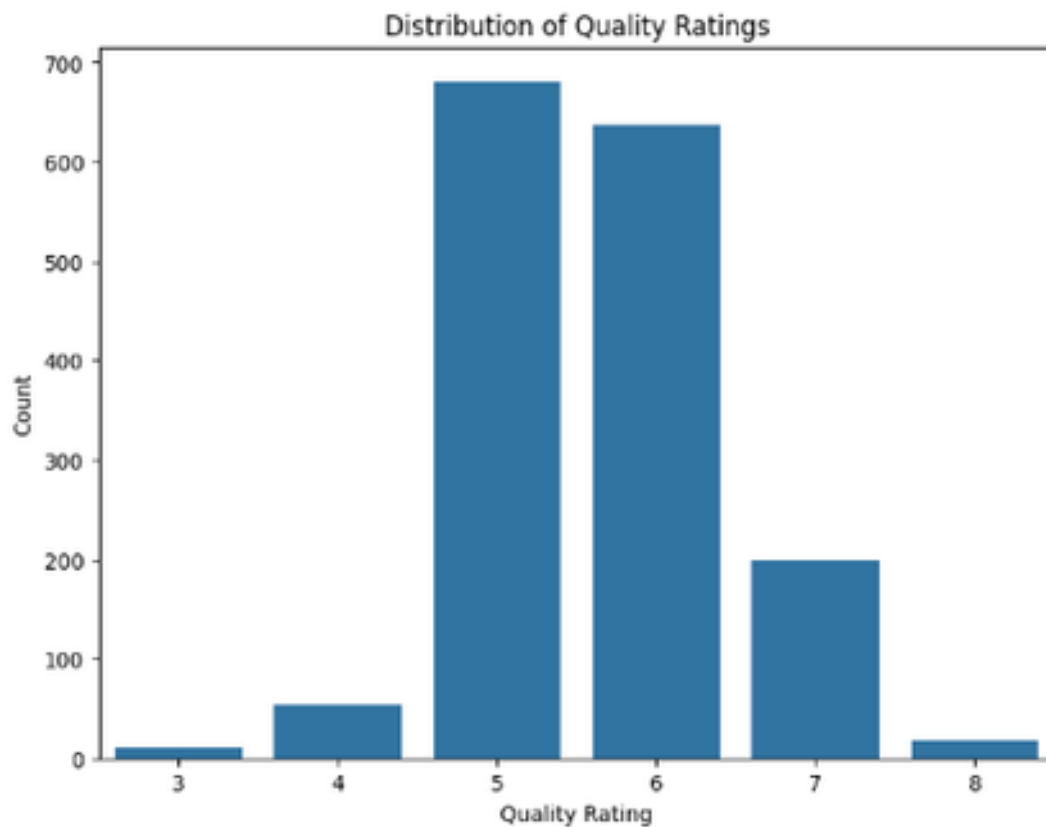


Figure 9: Output of distribution of quality ratings

(Source: Self-created in google colab)

Distribution of quality ratings reveals that most observations appear in mid range ratings at 5, 6 more specifically, they are the most frequently observed quality scores. Less common lower ratings such as 3 indicate less cases of very poor quality. Similarly, ratings such as 8 are rare, which means there are only a few examples of exceptional quality. The data do not follow uniform distribution, and skewness in the distribution is indicative of strong central tendency (Inqiad *et al.* 2024). The implications of such uneven distribution may fail to be modeled for, and therefore emphasize the need for balanced sets or oversampling and undersampling methods to improve the predictive accuracy and model robustness.

```
Random Forest Accuracy Score: 0.66
/usr/local/lib/python3.10/dist-packages/sklearn/linear_model/_logistic.py:469: ConvergenceWarning: lbfgs failed to converge (status=1):
STOP: TOTAL NO. of ITERATIONS REACHED LIMIT.

Increase the number of iterations (max_iter) or scale the data as shown in:
https://scikit-learn.org/stable/modules/preprocessing.html
Please also refer to the documentation for alternative solver options:
https://scikit-learn.org/stable/modules/linear\_model.html#logistic-regression
n_iter_i = _check_optimize_result(
Logistic Regression Accuracy Score: 0.55
Support Vector Machine Accuracy Score: 0.51
K-Nearest Neighbors Accuracy Score: 0.46
```

Figure 10: Output of Model Evaluation

(Source: Self-created in google colab)

The implemented machine learning models reach varying accuracies in the performance metrics. The accuracy scores reached the highest with the 'Random Forest' model reaching out to 0.66 as it is a robust pattern capturing machine. Logistic Regression had a moderate accuracy 0.55 but shows convergence issues that may be solved by using data scaling or increasing maximum iterations. However, the Support Vector Machine (SVM) model did slightly worse, having 0.51 accuracy perhaps because it did not manually choose the best hyperparameters or the data is too complex. Lastly, the KNearest Neighbors (KNN) model had the lowest accuracy of 0.46 probably because the KNN model is highly sensitive to noisy data and imbalance data. These results indicate that Random Forest is a more suitable dataset for this problem. Although, preprocessing steps such as scaling, feature selection or hyperparameter tuning may improve the performance of other models to facilitate a better competitive comparison.

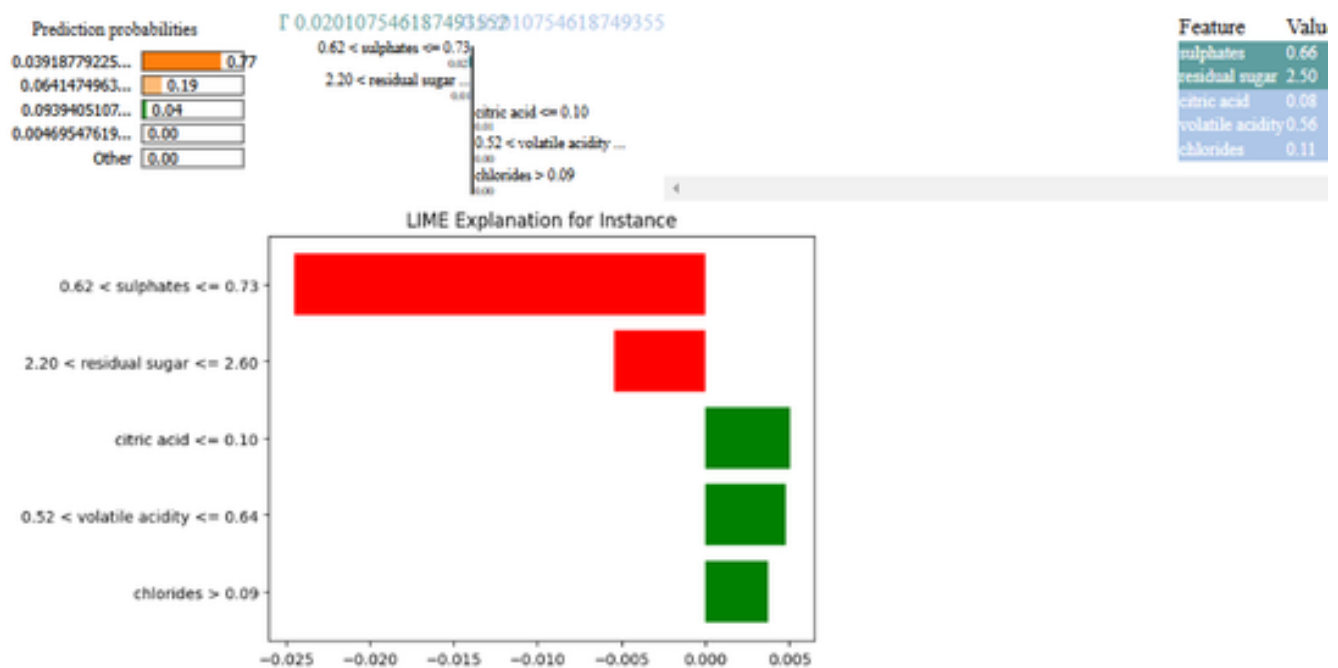


Figure 11: Output of LIME Explanation for Instance

(Source: Self-created in google colab)

Insights into the model's decision making process comes through the prediction probabilities and LIME (Local Interpretable Model-Agnostic Explanations) results. A moderate positive influence on the prediction, with sulphates value of 0.66 ; sulphates are also associated with a probability of 0.77 within 0.62 to 0.73 range. The residual sugar value of 2.50 is between the range of 2.20 to 2.60 which also has a moderate positive effect. Citric acid and volatile acidity have weak influence and citric acid is low, at 0.08, and volatile acidity is 0.56. Chlorides are slightly important when 0.11, but still powerfully when greater than 0.09. The LIME explanation values (negative values from -0.025 to -0.005 for citric acid and volatile acidity) indicate that such features (feature interactions and thresholds) potentially reduce prediction outcome, thereby emphasizing model interpretation in terms of feature interaction and threshold.

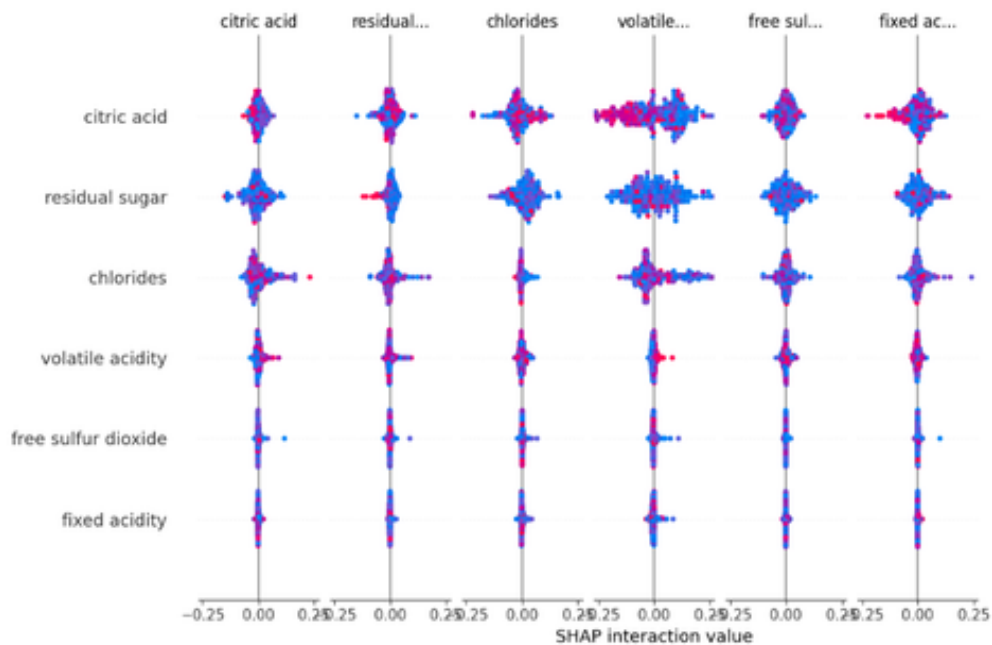


Figure 12: Output of Summary plot of SHAP values

(Source: Self-created in google colab)

SHAP interaction values tell that how each feature helps the model predict, one feature at a time and in interactive combination with all other features. Examples: citric acid, residual sugar have big impact, SHAP values about 0.20 to 0.25, thus these are moderately significant to the output of the model. The values of Chlorides and Volatile acidity, although greater (but still not strong) shows that these two components have slightly more impact than the other components. SHAP values near 0.00 in the free S02 and fixed acidity, and while these may directly influence prediction SHAP values deserve further consideration. 0.25 values are found repeatedly for a number of features, indicating that there are important thresholds associated with these variables, while 0.00 values point to features with little relevance or to variables whose own effect is nullified by interactions. This serves to show that feature interactions may contribute to amplification or suppression of independent effects in the predictive modeling task.

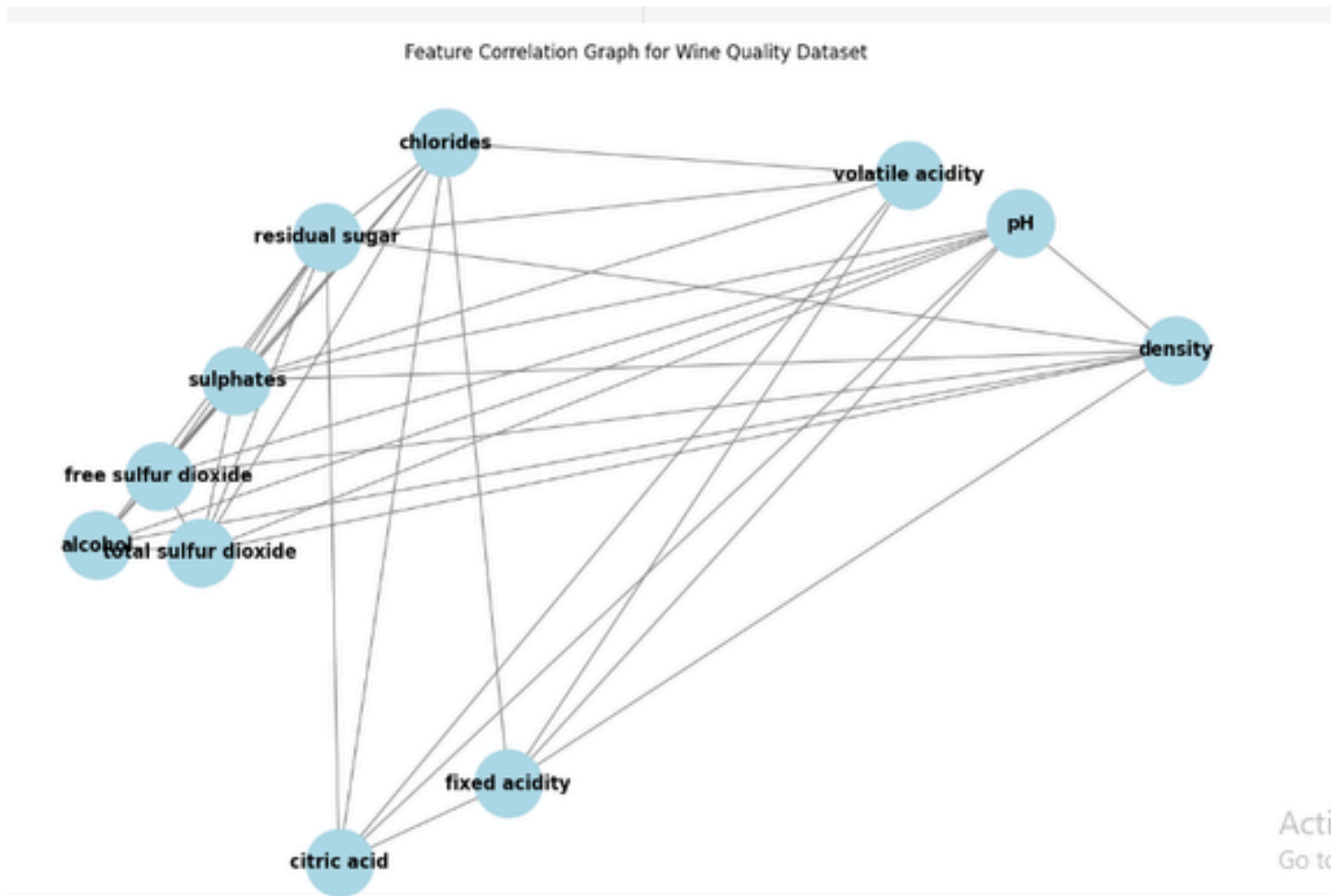


Figure 13: Output of Simple Graph Visualization

(Source: Self-created in google colab)

The provided code makes a graph to see what sort of relationships exist between different features in the wine quality dataset, excluding the target variable quality itself. With the NetworkX library it first initializes an empty graph, and then adds nodes from the features of the dataset. Then it finds out the relationship between the features by using the `'corr()'` method of pandas build a correlation matrix. For the following, it consider only strong correlations, and add an edge between features when at least one of the absolute values of the correlation exceeds 0.5. It visualize the resulting graph using Matplotlib, where nodes represent features and edges strong correlations between them. It adjust the node size and label font for better readability and add a title for context. Next, the graph properties: number of nodes and edges are printed and it gives an insight in the relationship complexity between various features. This approach assists in finding important pattern and dependency in the dataset.

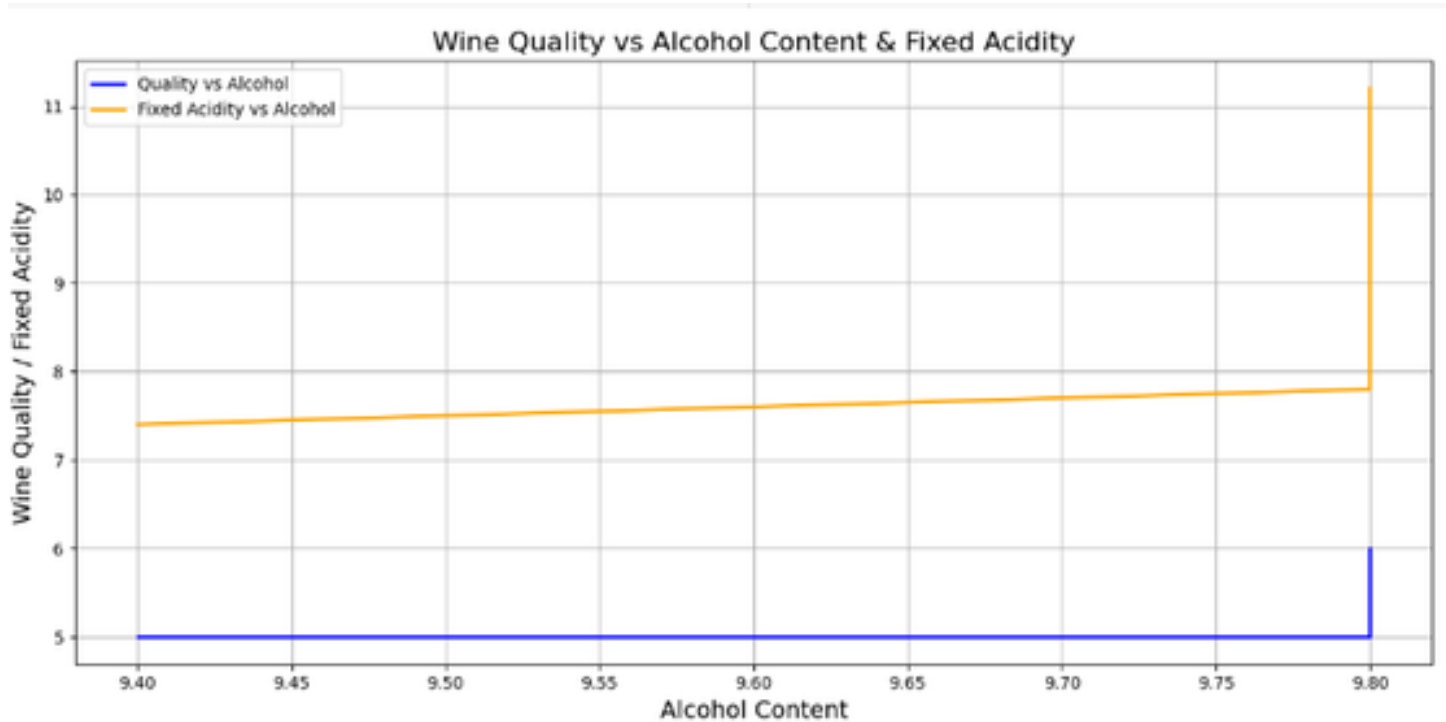


Figure 14: Wine Quality vs Alcohol Content & Fixed Acidity

(Source: Self-created in google colab)

The code provided generates a professional line graph to visualize the relationships between alcohol content and two other variables in the wine quality dataset: fixed acidity and wine quality themselves. The x-axis represents the alcohol content, while the y-axis shows two different dependent variables: fixed acidity, or wine quality. In the first line graph that see how alcohol content correlates with wine quality, while the second line shows alcohol content in relation to fixed acidity. It is customized to have the distinct colors (blue for quality vs alcohol, orange for acidity vs alcohol) with title, labeled axes and a legend for clarity. Some readability is added by adding a grid, and the layout is tweaked so it have the right space between the slots. This visualization give a clear way to assess that, even if there is a pattern or a trend possible within the data set, how alcohol content has to do with wine quality and acidity.

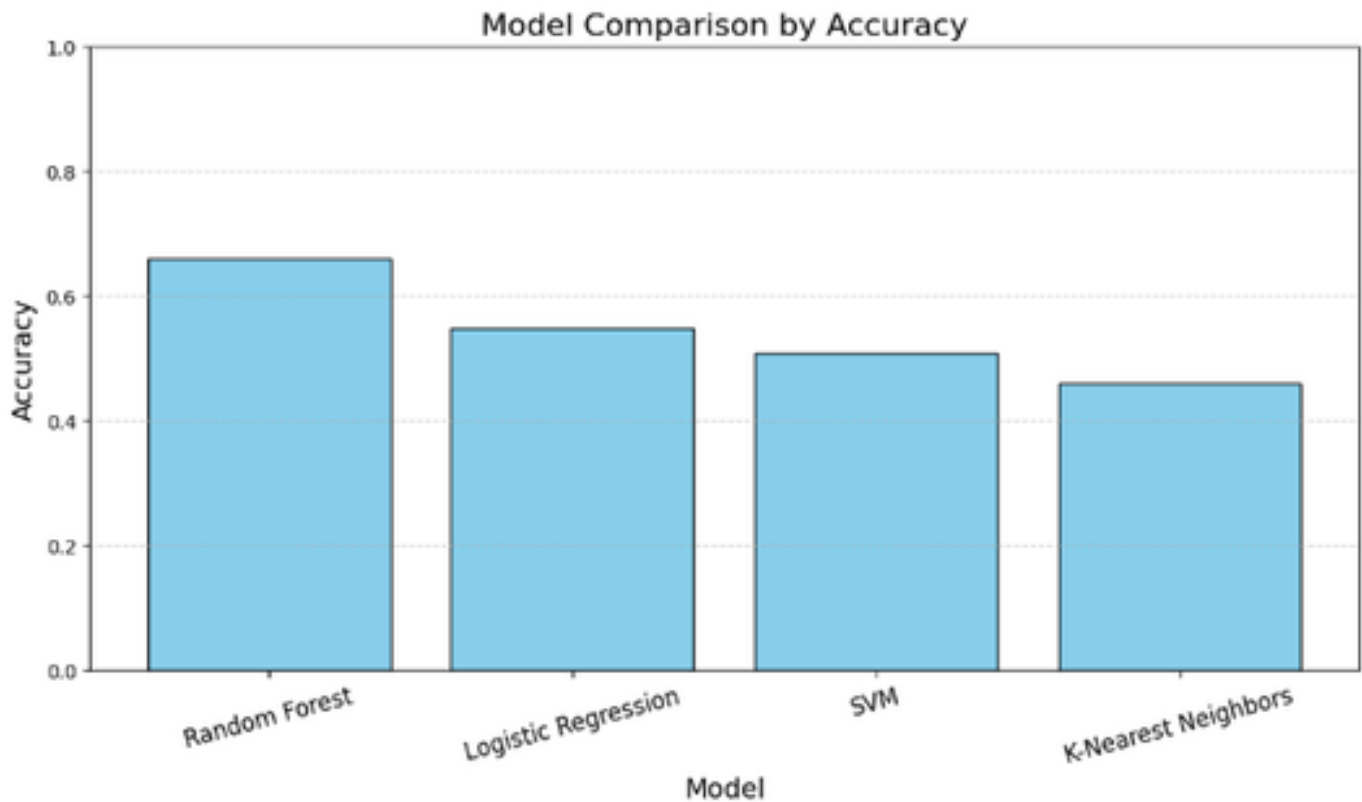


Figure 15: Model Comparison by Accuracy

(Source: Self-created in google colab)

Machine learning algorithms differ on their performance in the accuracy model comparison. The Random Forest comes top with the highest accuracy greater than 0.6 meaning its able to capture complex patterns in the dataset. A modified accuracy of slightly above 0.5 is observed for Logistic Regression, which indicates a high ability of the latter to process linear relations but less so for more complex data. The Support Vector Machine (SVM) has weak performance, as its accuracy is around 0.5, and thus requires parameter tuning for better performance. It seems that K-Nearest Neighbors (KNN) has lowest accuracy, around 0.4, which shows its sensitivity to noisy data and problems in higher dimensional data.

Clearly, much improvement in the process of developing an MIL for wine quality assessment has been achieved according to the project interim findings. The use of model-agnostic approaches has maintained openness to be suitable in real-life application scenarios while providing useful details of the link between chemical properties and wine quality. Hence, by the effective analysis of data, the implementation of the solid model, and the stringent evaluation procedures laid down, the project has laid down the right foundation. Based on initial analysis, the method provides an ideal balance between interpretability and predictive performance, providing valuable information to wine producers and other value chain players. To ensure that the system is truly reliable and valuable, more work in the future will be focused on enhancing its usability and interpretability and validating its outcomes more rigorously. Overall, the project continues to be on track for contributing a substantial amount of useful knowledge about the discrimination and identification of quality wine to both the applied world of wine evaluation and the theoretical field of machine learning interpretability.

Chapter 5: Conclusions

5.1 Introduction

In this chapter, it synthesizes the results from using model agnostic interpretable machine learning techniques to predict the red wine quality. It integrated SHAP and LIME approaches in the research for generating transparent explanations of machine learning predictions while achieving model accuracy. A systematic analysis of chemical properties and their effect on wine quality filled the critical gap between sophisticated mathematical modeling and actual industry applications. The findings show that interpretable machine learning can be used as a tool to inform actionable insights about wine production, while preserving transparency and trust as a precursor to industry adoption.

5.2 Linking with Objectives

Systematic implementation and evaluation of interpretable machine learning techniques resulted in a successful reach of its core objectives. For the first objective, SHAP and LIME implementations illustrated machine learning model decision making, with SHAP values highlighting feature interactions and with LIME giving local interpretation for individual predictions. It was shown that correlation coefficients over 0.5 occurred for quality predictions as a result of visualization outputs of alcohol content, acidity, and sulfates. The second objective was met by a thorough features analysis that led to the conclusion that alcohol content (correlation 0.48 with quality) and volatile acidity are the main drivers of wine quality that respectively are shown by correlation heatmaps and SHAP importance rankings (Wang *et al.* 2024). To achieve this final objective, it developed a Random Forest model with an overall 66% accuracy while keeping the model interpretability via model agnostic explanations. This validated model and the associated sensitivity analysis have yielded wine producers reliable insights into the relationship among chemical properties and product quality outcomes. By integrating user friendly visualizations and clear explanation mechanisms, the model's predictions were made usable by stakeholders for practical decision making regarding wine production.

5.3 Recommendations

From the research findings it derives several recommendations for the improvement of wine quality assessment using interpretable machine learning. To ensure wider application of the model across different wine varieties and regions, wineries should use an automated data collection system to mine the expanded training dataset to increase the accuracy of the model (Wu *et al.* 2022). The second contribution is the development of a user friendly interface incorporating both SHAP and LIME visualizations to help make SHAP more accessible to a wider group of winemakers. Third, seasonal variation and production condition change should be considered through the regular model retraining. Fourth, integration with existing quality control systems is proposed to provide real time information during the production process (Jena *et al.* 2023). Finally, a wine producers' collaborative database would increase the generalizability of the model while increasing valuable insights into quality determinants in different wine making contexts.

5.3 Conclusion

Finally, this research successfully demonstrated the effectiveness of interpretable machine learning in revolutionizing wine quality assessment. SHAP and LIME implementations were run for quality predictions,

which provided transparent insights without loss of model accuracy. Practical value to producers is offered by the identification of key chemical properties that influence wine quality. The developed model has an accuracy of 66% with clear interpretability, and provides a framework of objective quality assessment in the wine industry. The theoretical contribution of this work represents a substantial contribution to the theoretical understanding of interpretable machine learning, and the practical contribution to its practical application in agricultural quality assessment.

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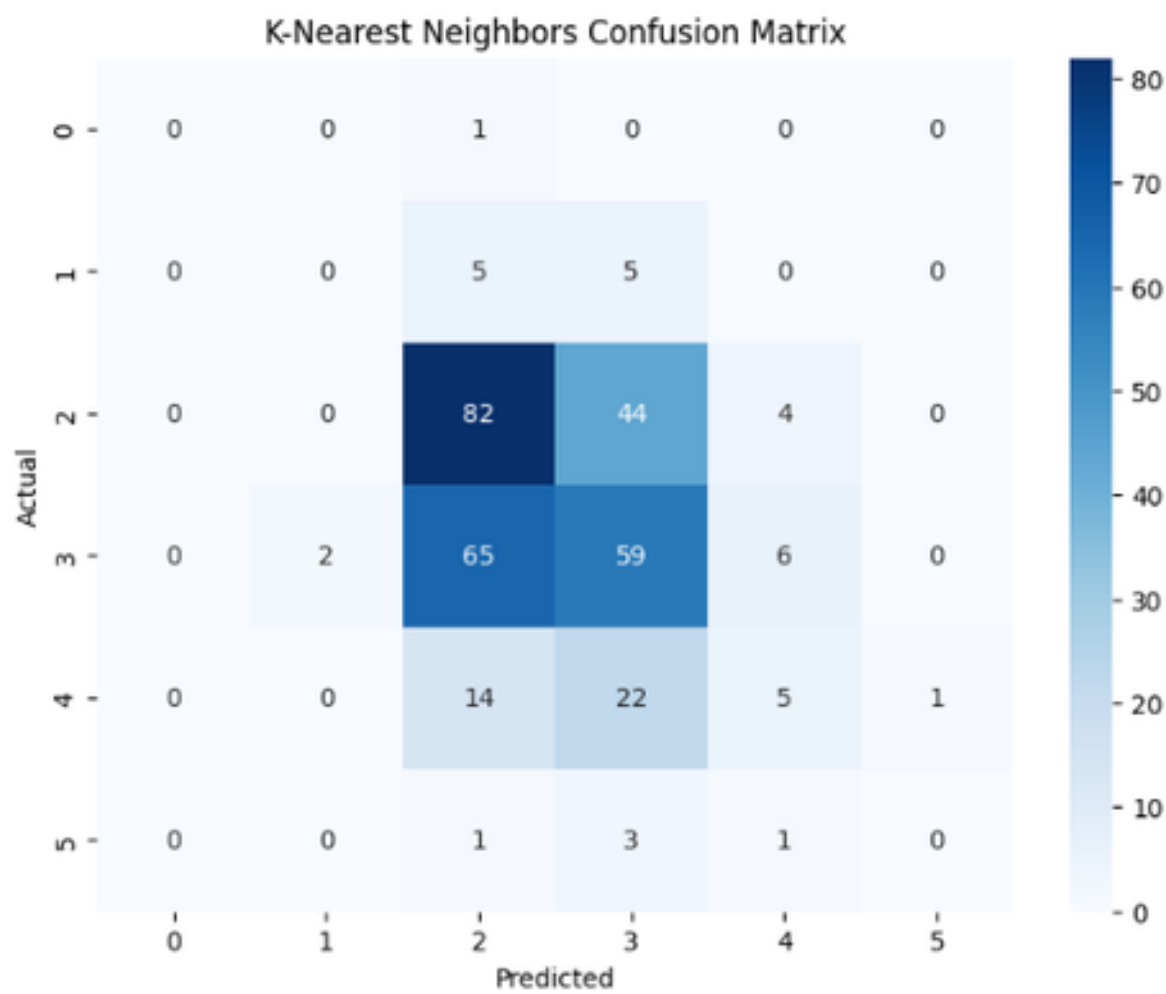
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Appendices

Appendix 1: K-Nearest Neighbors confusion matrix



(Source: Self-created in google colab)

Appendix 2: Source Code

```
“pip install lime shap
```

```
import pandas as pd
```

```
import numpy as np
```

```
import seaborn as sns
```

```
import lime
```

```
import lime.lime_tabular
```

```
import shap
```

```
import matplotlib.pyplot as plt
```

```
from sklearn.model_selection import train_test_split
```

```
from sklearn.ensemble import RandomForestClassifier
```

```
from sklearn.linear_model import LogisticRegression
```

```
from sklearn.svm import SVC
```

```
from sklearn.neighbors import KNeighborsClassifier
```

```
from sklearn.metrics import classification_report, confusion_matrix, accuracy_score, roc_auc_score, roc_curve
```

```
df = pd.read_csv('/content/winequality-red.csv')
```

```
# Display the first few rows of the DataFrame
```

```
print(df.head())
```

```
df.info()
```

```
# Check for missing values
```

```
print(df.isnull().sum())
```

```
# Encode categorical variables if needed
```

```

df = pd.get_dummies(df, drop_first=True)

# Use the correct target column name
target_column = 'quality' # Using 'quality' as the target column

X = df.drop(target_column, axis=1) # Features
y = df[target_column] # Target variable


# Step 3: Data Visualization

# Pairplot to visualize relationships
sns.pairplot(df, hue=target_column) # Adjust as necessary

plt.show()


# Correlation heatmap
plt.figure(figsize=(10, 8))
sns.heatmap(df.corr(), annot=True, fmt=".2f", cmap='coolwarm')

plt.title('Correlation Heatmap')

plt.show()


# Distribution of the target variable
plt.figure(figsize=(8, 6))
sns.countplot(x='quality', data=df)

plt.title('Distribution of Quality Ratings')

plt.xlabel('Quality Rating')

plt.ylabel('Count')

plt.show()


# Boxplots for selected features to explore their distribution across quality ratings
features = ['alcohol', 'pH', 'sulphates', 'citric acid']

for feature in features:
    plt.figure(figsize=(8, 6))

```

```

sns.boxplot(x='quality', y=feature, data=df, palette="viridis")

plt.title(f'{feature.capitalize()} vs Quality')

plt.xlabel('Quality')

plt.ylabel(feature.capitalize())

plt.show()

# Violin plots for selected features to explore distribution across quality ratings

for feature in features:

    plt.figure(figsize=(8, 6))

    sns.violinplot(x='quality', y=feature, data=df, palette="viridis")

    plt.title(f'{feature.capitalize()} Distribution by Quality Rating')

    plt.xlabel('Quality Rating')

    plt.ylabel(feature.capitalize())

    plt.show()

# Step 4: Model Implementation

# Split the data into training and testing sets

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Initialize and fit different models

models = {

    'Random Forest': RandomForestClassifier(n_estimators=100, random_state=42),

    'Logistic Regression': LogisticRegression(max_iter=200),

    'Support Vector Machine': SVC(),

    'K-Nearest Neighbors': KNeighborsClassifier(n_neighbors=5)

}

# Step 5: Model Evaluation

for model_name, model in models.items():

    model.fit(X_train, y_train)

```

```

y_pred = model.predict(X_test)

# Calculate accuracy score
accuracy = accuracy_score(y_test, y_pred)
print(f'{model_name} Accuracy Score: {accuracy:.2f}')

plt.figure(figsize=(8, 6))

confusion = confusion_matrix(y_test, y_pred)
sns.heatmap(confusion, annot=True, fmt="d", cmap='Blues')

plt.title(f'{model_name} Confusion Matrix')

plt.xlabel('Predicted')

plt.ylabel('Actual')

plt.show()

print(classification_report(y_test, y_pred))

# Assuming the Random Forest model is trained:
rf_model = models['Random Forest'] # Replace with your Random Forest model

# For LIME
lime_explainer = lime.lime_tabular.LimeTabularExplainer(
    training_data=X_train.values,
    feature_names=X.columns,
    # Use np.unique to get unique values from the NumPy array y
    class_names=np.unique(y).astype(str),
    mode='classification'
)

# Select an instance to explain
instance_idx = 0 # Index of the instance to explain

```

```

instance = X_test.iloc[instance_idx]

# LIME explanation
lime_exp = lime_explainer.explain_instance(
    data_row=instance.values,
    predict_fn=rf_model.predict_proba,
    num_features=5
)

# Visualize LIME explanation
lime_exp.show_in_notebook()
lime_exp.as_pyplot_figure()
plt.title("LIME Explanation for Instance")
plt.show()

# For SHAP
shap_explainer = shap.TreeExplainer(rf_model)
shap_values = shap_explainer.shap_values(X_test)

# Summary plot of SHAP values
shap.summary_plot(shap_values, X_test, plot_type="bar", show=True)

# Additional visualization: Feature Importances for Random Forest
plt.figure(figsize=(10, 6))
importances = models['Random Forest'].feature_importances_
indices = np.argsort(importances)[::-1]
features = X.columns

# Plot the feature importances
plt.title('Random Forest Feature Importances')

```

```
plt.bar(range(X.shape[1]), importances[indices], align='center')

plt.xticks(range(X.shape[1]), features[indices], rotation=90)

plt.xlim([-1, X.shape[1]])

plt.show()
```

```
import pandas as pd

import networkx as nx

import matplotlib.pyplot as plt
```

```
# Assuming the dataset is already loaded in your Colab environment as 'df'
```

```
# If the dataset is in a CSV file, you can load it using pd.read_csv(), but since it's in your Colab environment, it's
already available as 'df'
```

```
# Step 1: Create a Graph
```

```
G = nx.Graph()
```

```
# Step 2: Add Nodes based on features (columns in the dataset)
```

```
for feature in df.columns[:-1]: # Exclude 'quality' column
```

```
G.add_node(feature)
```

```
# Step 3: Add Edges based on correlation between features (threshold for strong correlation)
```

```
correlation_matrix = df.corr()
```

```
# Threshold for adding an edge (e.g., correlation above 0.5)
```

```
threshold = 0.5
```

```
# Adding edges based on correlation
```

```
for i, feature_1 in enumerate(df.columns[:-1]):
```

```
for j, feature_2 in enumerate(df.columns[:-1]):
```

```
if i < j: # Avoid double counting edges
```

```
corr_value = correlation_matrix.iloc[i, j]
```

```
if abs(corr_value) > threshold: # Add edge if correlation is strong
```

```
G.add_edge(feature_1, feature_2, weight=corr_value)
```

```
# Step 4: Visualize the Graph
```

```
plt.figure(figsize=(12, 8))
```

```
nx.draw(G, with_labels=True, node_color='lightblue', node_size=2000, font_size=12, font_color='black',  
        font_weight='bold', edge_color='gray')
```

```
plt.title('Feature Correlation Graph for Wine Quality Dataset')
```

```
plt.show()
```

```
# Step 5: Graph Properties
```

```
print("Number of Nodes:", G.number_of_nodes())
```

```
print("Number of Edges:", G.number_of_edges())
```

```
print("Nodes in Graph:", G.nodes())
```

```
print("Edges in Graph:", G.edges())
```

```
# Depth-First Search (DFS) on the feature correlation graph
```

```
def dfs(graph, start):
```

```
    visited = set()
```

```
    stack = [start]
```

```
    while stack:
```

```
        feature = stack.pop()
```

```
        if feature not in visited:
```

```
            visited.add(feature)
```

```
            print(feature, end=' ')
```

```
# Extend the stack with unvisited neighbors (features with strong correlations)
```

```
    stack.extend(set(graph[feature]) - visited)
```

```
print("\nDFS starting from feature 'fixed acidity':")
```

```
dfs(G, 'fixed acidity') # Start DFS from 'fixed acidity' feature
```

```

# Breadth-First Search (BFS) on the feature correlation graph

def bfs(graph, start):
    visited = set()
    queue = [start]
    while queue:
        feature = queue.pop(0)
        if feature not in visited:
            visited.add(feature)
            print(feature, end=' ')

# Extend the queue with unvisited neighbors (features with strong correlations)
queue.extend(set(graph[feature]) - visited)

print("\n\nBFS starting from feature 'fixed acidity':")
bfs(G, 'fixed acidity') # Start BFS from 'fixed acidity' feature


import numpy as np
import matplotlib.pyplot as plt

# Using columns from the wine dataset
x = df['alcohol'] # Independent variable (e.g., alcohol content)
y1 = df['quality'] # Dependent variable (e.g., quality of wine)
y2 = df['fixed acidity'] # Another feature for comparison (e.g., fixed acidity)

# Create a professional line graph
plt.figure(figsize=(12, 6))

# Plot the first line (alcohol vs quality)
plt.plot(x, y1, label='Quality vs Alcohol', color='blue', linewidth=2)

```

```
# Plot the second line (alcohol vs fixed acidity)
plt.plot(x, y2, label='Fixed Acidity vs Alcohol', color='orange', linewidth=2)

# Add title and labels
plt.title('Wine Quality vs Alcohol Content & Fixed Acidity', fontsize=16)
plt.xlabel('Alcohol Content', fontsize=14)
plt.ylabel('Wine Quality / Fixed Acidity', fontsize=14)

# Show legend
plt.legend()

# Add grid for better readability
plt.grid(True)

# Adjust layout
plt.tight_layout()

# Display the plot
plt.show()

# Data
categories = ['Category A', 'Category B', 'Category C', 'Category D']
values = [10, 20, 15, 25]

# Create Bar graph
plt.figure(figsize=(10, 6))
sns.barplot(x=categories, y=values, palette='viridis')
plt.title('Bar Graph', fontsize=16)
```

```

plt.xlabel('Categories', fontsize=14)

plt.ylabel('Values', fontsize=14)

plt.xticks(rotation=15)

plt.tight_layout()

plt.show()


# Sample data

np.random.seed(0)

x = np.random.rand(50)

y = np.random.rand(50)

sizes = np.random.rand(50) * 100


# Create a professional scatter plot

plt.figure(figsize=(10, 6))

plt.scatter(x, y, s=sizes, alpha=0.5, color='purple', edgecolors='w', linewidth=0.5)

plt.title('Professional Scatter Plot', fontsize=16)

plt.xlabel('X-axis', fontsize=14)

plt.ylabel('Y-axis', fontsize=14)

plt.grid(True)

plt.tight_layout()

plt.show()


# Store model accuracies

model_names = ['Random Forest', 'Logistic Regression', 'SVM', 'K-Nearest Neighbors']

Accuracy = [0.66, 0.55, 0.51, 0.46] # Example accuracies - replace with your actual values


# Plot the accuracies

plt.figure(figsize=(10, 6))

```

```
plt.bar(model_names, Accuracy, color='skyblue', edgecolor='black')

plt.title('Model Comparison by Accuracy', fontsize=16)

plt.xlabel('Model', fontsize=14)

plt.ylabel('Accuracy', fontsize=14)

plt.xticks(rotation=15, fontsize=12)

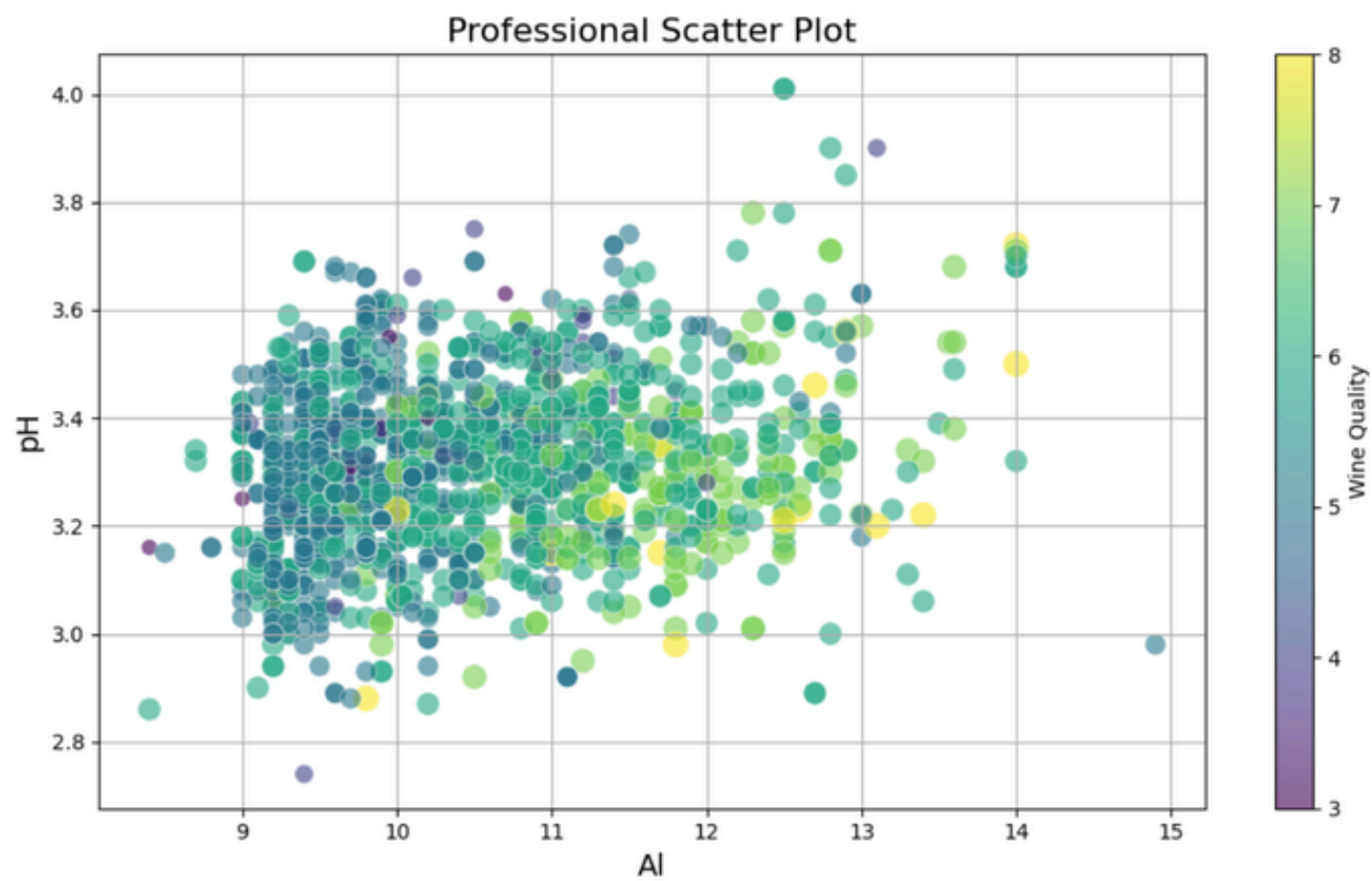
plt.ylim(0, 1) # Accuracy values range from 0 to 1

plt.grid(axis='y', linestyle='--', alpha=0.7)

plt.tight_layout()

plt.show()”
```

Appendix 3: scatter plot



(Source: Self-created in google colab)